

Consolidated provisional materials disclosure

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Scope and remaining facts

The requested scope remains all nonempty supports through 138 elemental labels, all normalized rational targets including arbitrarily small positive fractions, symbolic real domains, isotope/state and particle constructions, and expected/contrary property scenarios. The 118 recognized labels and 20 hypothetical placeholders retain separate status. Numerical descriptions and dated identities do not establish arbitrary preparation, properties or future legal effect.

The original 80 numbered paragraphs and 199 candidate claims are preserved in Part A. Its original printed page numbers and preparation date remain unchanged; PDF labels and this map provide consolidated navigation. The two drawings appear on Part A pages 10-11. Parts B/C print the actual new technical and literature text. Part D qualifies earlier publication-stage statements. Part E retains the complete requested objective and its unresolved evidence. Part F maps every candidate to actual original pages.

The separate companion materials_identity_data_review_annex.pdf has 795 review pages and SHA-256 a808672808c7861f03b94a06771996b5fc8f94973b94612b131a1df37ba8fd. Its intended scope is the complete scoped NUBASE ASCII rows and PDG identity selections with schema/provenance; validate its own source counts and values using the annex tool and report. Those full record sets are NOT embedded in this core PDF; include and validate the companion if relying on them. No full PDG property database or unrestricted current/future entity universe is included. External links identify sources and do not replace absent essential technical content. Review page counts are not a PCT chargeable-sheet or accepted-filing-format determination.

TECH-001 remains open: identify the actual human technical contribution and responsible inventors. Complete source-critical preparation gaps, selected embodiment support, claim-specific patent analysis and actual applicant/entitlement information before calling a packet ready to submit. No applicant facts or experiments are invented. The original filing memorandum and earlier PDFs remain separate historical review artifacts; their old page/check counts do not certify this edition.

Included content and navigation

Part	Included content	Physical PDF pages
A	Original application working draft (29 September 2026) (27 pages)	3-29
B	Technical definitions and accounting (6 pages)	30-35
C	Five material families and proposed study cards (13 pages)	36-48
D	Current publication and filing status (1 pages)	49-49
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F	Per-claim page and textual-support map (11 pages)	56-66

Exact source versions

These hashes identify read-only build inputs. They are not trusted timestamps or proof of earliest publication.

Included source file	SHA-256
documents/pdf/provisional_application_draft.pdf	a8306290a92c9433271dd1711e1bdabfd1c79779877897c99186638f3d6e0d7f
documents/provisional_application_draft.md	3c3a6f9b42ca87ffedfd97435ac0476d3301f7f1d650b4c2b9f0fb0b66b15949
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documents/drawings/material_record_sequence.svg	9b28132a377a157509ca7f8885978a59d89f071719125fe6f761772269650c55

Draft provisional materials disclosure

Prepared 29 September 2026 | Technical working draft for applicant and patent counsel

This draft develops the requested all-element composition framework, including single-element materials, every component count from 2 through 138, exact ranges, isotope parameters, particle categories, and conditional property scenarios. The numbered description and candidate claim annex are intended to be reviewed together. No application has been filed. No applicant laboratory results, identified inventive contribution, or full-scope enablement determination has been supplied. Mathematical definitions are exact; they are not evidence that every defined material can be prepared.

1 Technical field

[0001] The subject matter concerns the description, preparation, characterization, and comparison of materials having specified elemental and isotopic compositions, structures, processing histories, and properties. Materials considered include metals and alloys, inorganic compounds and ceramics, molecular substances, polymers, composites, mixtures, and single-element forms. Particle-containing or excited systems are addressed separately from ordinary chemical constituents.

[0002] A composition framework is provided to identify selected constituents, normalize their proportions, express exact values and intervals, and associate each numerical description with a structural and preparation record. Its purpose is to make the requested scope explicit and reproducible as a specification of candidate compositions. Whether an individual candidate exists, can be synthesized, or has a useful property requires additional technical information.

2 Background and evidence status

[0003] An elemental formula does not uniquely define a material. Connectivity, crystal structure, phase assemblage, isotope distribution, defects, interfaces, orientation, morphology, and processing can distinguish systems sharing the same elemental proportions. A molecular formula can describe different isomers; a ceramic formula can describe different polymorphs; a polymer formula can omit sequence and architecture; a composite's overall percentages can omit its load-bearing or conducting network.

[0004] The element vocabulary has two groups. The recognized group consists of atomic numbers 1 through 118 in the IUPAC periodic table. The requested extension consists of unconfirmed placeholders with atomic numbers 119 through 138. The extension is an indexing convention for hypothetical candidates, not a statement that these elements have been discovered or possess known chemistry. [IUPAC periodic table](#)

[0005] Four evidence labels are used locally: ESTABLISHED for an attributed source teaching; CALCULATED for a stated mathematical result or model calculation; PROPOSED for an unperformed preparation or test with specified assumptions; and UNSUPPORTED for a possibility without adequate preparation or property support. These labels do not decide patentability. A calculated fraction is not a measured composition, and a proposed recipe is not a performed experiment.

[0006] No new synthesis, measured yield, verified phase, measured property, or applicant-specific successful embodiment is reported in this draft. Published third-party examples retain their attribution. The requested breadth includes unsupported candidates. These candidates are expressly identified so that the scope is visible without representing them as established teachings.

3 Summary of the composition framework

[0007] A candidate material is described by a record $R = (B, S, x, I, U, C, P, M, F, E)$, where B is the composition basis; S is the constituent support set; x is the normalized composition vector; I is the isotope and nuclear-state description; U is the structural and spatial description; C is the preparation and conditioning record; P is the property scenario; M is the measurement record; F identifies failure boundaries; and E identifies evidence and sources. Missing fields are stated explicitly.

[0008] The framework includes every nonempty subset of the 118 recognized element labels and, separately, every nonempty subset of the 138-label vocabulary including the hypothetical extension. It includes every positive normalized continuous composition as a target description and every rational normalized composition as a separate exact count-compatible description. A finite decimal grid is a useful additional subset, not a replacement for either domain.

[0009] Each interval embodiment is the intersection of its component bounds with the normalization condition. Component endpoints cannot be selected independently if their sum would differ from one. Numerical enumeration provides a reproducible address for a candidate; it supplies neither missing chemistry nor a proof that later claims to that candidate lack novelty.

4 Composition bases and constituent accounting

[0010] On an atomic basis, $x_i = N_i / \sum(N_j)$, where N_i counts atoms of element i in the specified inventory. A mole-of-atoms basis has the same normalized proportions when the same atom inventory is used. A mole-of-molecules, formula-unit, repeat-unit, site-occupancy, or feed-component basis is different and must be expressly named. Denominators must identify the counted entity, sample region, and inclusion of impurities.

[0011] On a mass basis, $w_i = m_i / \sum(m_j)$. Conversion from atomic fractions uses $w_i = x_i M_i / \sum(x_j M_j)$, with M_i appropriate to the specified isotope distribution and units. The inverse uses $x_i = (w_i / M_i) / \sum(w_j / M_j)$. Standard atomic weights are not substitutes for unknown masses of hypothetical nuclei. Conversion is unavailable where the needed mass information is unavailable.

[0012] Volume fractions describe identified phases or constituents under specified temperature, pressure, density, and volume-assignment conventions. They are not automatically elemental atomic fractions. Mixing volumes need not be additive. A composite record therefore identifies constituent densities and whether volume fractions are measured, calculated from a justified additive model, or nominal feed values.

[0013] Feed proportions, retained product composition, and local measured composition are separate fields. A specimen may contain gradients, layers, inclusions, or surface enrichment. The support set identifies which constituents have positive fractions in the stated denominator. Zero entries are omitted from the canonical support set, while a zero concentration endpoint can describe a boundary shared with a smaller support set.

[0014] A closed composition specification includes all constituents in its denominator. An open specification can permit additional constituents, but must state how their inclusion affects normalization. A record describing selected constituents relative only to their subtotal is not a complete bulk-composition assay. Residual impurities are included as identified amounts, bounded unknown residuals, or an expressly unresolved term; they are not silently set to zero.

5 Exact constituent support and composition domains

[0015] Let $E = \{E_1, \dots, E_{118}\}$, with E_Z corresponding to the recognized element of atomic number Z . Let $H = \{E_{119}, \dots, E_{138}\}$ denote the hypothetical placeholders. For either vocabulary V , choose an ordered support $S = (E_{z1}, \dots, E_{zk})$, with strictly increasing atomic numbers. The ordering removes duplicate permutations; it does not restrict spatial ordering within a material.

[0016] For each integer k from 1 through the vocabulary size, the continuous target domain is $\Delta(S) = \{x : x_i > 0 \text{ for every member of } S, \sum(x_i) = 1\}$. Its closure permits $x_i = 0$, with the resulting candidate reassigned to its positive support. Every value is dimensionless; percent values equal $100 x_i$. For $k = 1$, the only normalized vector is (1).

[0017] The rational domain consists of all $x_i = n_i/q$, where q is any positive integer, each n_i is a positive integer, and $\sum(n_i) = q$. For k constituents, q must be at least k . A primitive representation has $\gcd(q, n_1, \dots, n_k) = 1$. This removes duplicate representations such as (1/2, 1/2) and (2/4, 2/4). The unrestricted denominator includes (1/3, 2/3), which is absent from a grid whose denominator is 10^{19} .

[0018] The requested decimal count grid has $D = 10^{19}$. Its vectors satisfy $x_i = n_i/D$, $n_i \geq 1$, and $\sum(n_i) = D$. The smallest positive grid fraction is $a = 1/10^{19} = 0.0000000000000000001$. In percent units it is 0.000000000000000001% . This is a chosen grid increment, not a universal physical concentration limit.

[0019] For k positive constituents on this grid, no single constituent can exceed $1 - (k-1)/D$. A two-component extreme is $(1/D, (D-1)/D)$. A 138-component extreme has 137 constituents each at $1/D$ and the remaining constituent at $(D-137)/D$. These vectors normalize exactly. They do not establish that a chemically homogeneous or stable 138-element material can be made.

[0020] The broad continuous and unrestricted rational domains remain available below the selected grid floor. If a particular embodiment imposes a positive floor a , then $ka \leq 1$ is necessary and sufficient for an unconstrained normalized vector to exist. For a fixed total atom inventory N , the least positive atomic count fraction is $1/N$. Since N depends on the sample, no sample-independent lower fraction follows from atom counting.

[0021] The SI Avogadro constant is exactly $6.02214076 \times 10^{23}$ per mole. An inventory of 10^{19} atoms is approximately 0.00001660539067 mole of atoms. One specified atom among that inventory gives a count fraction of 10^{-19} . The arithmetic does not establish controlled placement, detectability, representative sampling, or a reproducible bulk property. [BIPM definition of the mole](#)

[0022] The recognized vocabulary has $2^{118} - 1$ nonempty supports. Its supports of size at least two number $2^{118} - 1 - 118$. The extended vocabulary has $2^{138} - 1$ nonempty supports and $2^{138} - 1 - 138$ supports of size at least two. For a fixed selected support of size k on the D grid, the number of positive vectors is $\text{binomial}(D-1, k-1)$. These are counts of mathematical descriptions, not counts of enabled substances.

6 All endpoint ranges and complementary proportions

[0023] For each precision index $p = 0, 1, \dots, 19$, let $Q_p = 10^p$. Select integers l and u with $0 \leq l \leq u \leq Q_p$. The closed interval $I(p, l, u) = [l/Q_p, u/Q_p]$ defines every pair of decimal-grid endpoints at that precision, including exact-point intervals $l = u$. Percent endpoints are $[100 l/Q_p, 100 u/Q_p]$. Every elementary cell $[j/Q_p, (j+1)/Q_p]$ is included for $0 \leq j < Q_p$, as is every union represented by a contiguous pair of endpoints.

[0024] At $p = 2$ the elementary cells are $[0\%, 1\%]$, $[1\%, 2\%]$, ..., $[99\%, 100\%]$. For a binary composition, if x_A lies in $[L, U]$, $x_B = 1 - x_A$ lies in $[1-U, 1-L]$. Thus $[1\%, 2\%]$ pairs with $[98\%, 99\%]$, $[2\%, 3\%]$ with $[97\%, 98\%]$, and $[49\%, 50\%]$ with $[50\%, 51\%]$. Corresponding endpoints describe correlated compositions: (1%, 99%) and (2%, 98%), not (1%, 98%).

[0025] Closed cells share their endpoints. This intentional overlap avoids losing boundary values. Canonical indexing may assign a shared boundary to one cell for retrieval, but the physical composition is unchanged. Exact boundary points, every intermediate target value, each contiguous subrange with specified endpoints, and each complement are contemplated as numerical descriptions. This language does not assert that all such subranges are legally disclosed with sufficient specificity.

[0026] The requested interval 0.0000000000000001% through 0.0000000000000001% equals $[10^{-18}, 10^{-17}]$ as a fraction. On the D grid its numerators are 10 through 100 inclusive: 91 exact grid values and 90 elementary bins. The binary complement is $[(D-100)/D, (D-10)/D]$, or $[99.99999999999999\%, 99.99999999999999\%]$. Fraction and percent units must not be interchanged.

[0027] The 50/50 binary vector is exactly $(1/2, 1/2)$. The entire decimal hierarchy connects ordinary percent cells to the requested very small cells. Additional rational intervals $[r, s]$, for all rational $0 \leq r \leq s \leq 1$, and real endpoint target intervals may be specified outside the decimal hierarchy. Real targets, rational finite-count descriptions, decimal grid values, and measured tolerances remain distinct.

[0028] A decade refinement supplies optional addresses for very small positive values. For e from -19 through -1 and any nonnegative integer r , define intervals $[j \times 10^{(e-r)}, (j+1) \times 10^{(e-r)}]$ with $j = 10^r, \dots, 10^{(r+1)}-1$. These partition each selected decade at the specified refinement, with shared closed boundaries. Refinement beyond the D grid is symbolic or rational coverage, not an assertion of greater attainable manufacturing precision.

[0029] For a selected k -component support choose an interval $[L_i, U_i]$ for every component, with $0 \leq L_i \leq U_i \leq 1$, after intersecting any supplied bounds with $[0, 1]$. The candidate domain is $\{x : L_i \leq x_i \leq U_i, \sum(x_i) = 1\}$, intersected with the applicable positivity or grid conditions. A box with closed component bounds is feasible on the nonnegative continuous simplex exactly when each $L_i \leq U_i$ and $\sum(L_i) \leq 1 \leq \sum(U_i)$. This test is applied after intersecting with any selected positive floor and corresponding upper bound.

[0030] A witness can be constructed exactly: start with $x_i = L_i$; let residual $r = 1 - \sum(L_i)$; in support order add $\min(r, U_i - L_i)$ to each component and reduce r accordingly. The box is feasible if the residual reaches zero. With rational inputs the calculation uses integer fractions. Strict positivity without a fixed floor needs a separate check; a greedy witness containing zero does not prove that a positive witness is unavailable. With the ordinary feasibility conditions satisfied, require every $U_i > 0$. If $\sum(L_i) = 1$, feasibility with this support additionally requires every $L_i > 0$. If $\sum(L_i) < 1$, choose $\epsilon = \min(\min_i U_i, (1 - \sum(L_i))/(2k))$, replace each lower bound by $\max(L_i, \epsilon)$, and perform the residual allocation. ϵ is positive and the raised lower-bound sum remains below one, so the construction returns a positive witness. With rational inputs all operations remain exact. A particular vector having zero entries is reassigned to its smaller positive support.

[0031] For the D grid, set $b_i = \max(1, \text{ceiling}(D L_i))$ and $c_i = \min(D-k+1, \text{floor}(D U_i))$, and test $b_i \leq c_i$ and $\sum(b_i) \leq D \leq \sum(c_i)$. The same residual allocation uses integers. A box may be continuously feasible and contain no D-grid vector. For example the exact binary vector $(1/3, 2/3)$ is rationally feasible but is not on the D grid.

[0032] Independent selection of all endpoint ranges is permitted as an indexing operation, but infeasible combinations are retained only as rejected numerical requests, with the reason recorded. Example: three lower bounds of 40% cannot normalize. A valid ternary example has atomic intervals $A = [10\%, 20\%]$, $B = [30\%, 40\%]$, $C = [40\%, 60\%]$, with witness $(10\%, 30\%, 60\%)$. No product phase or preparation follows from that witness.

7 Structural and spatial descriptions

[0033] A material record identifies whether the composition describes a single phase, multiple phases, a molecular species, a solution, a dispersed mixture, a layered article, or an ensemble. Where the distinction is unknown, it is marked unknown. A numerical vector does not imply every constituent occupies a common lattice or remains uniformly distributed.

[0034] For a crystalline candidate, specify the phase identity, space group or structural model where known, lattice parameters with conditions, occupied sites, substitutions, vacancy rules, ordering, and charge-balance constraints. A proposed site assignment is labeled PROPOSED. Unknown structure is not filled by saying all crystal structures are present.

[0035] For an amorphous candidate, identify the preparation history, detectable crystalline fraction, relevant local-order evidence, glass transition or relaxation behavior where measured, and the test's resolution. For molecular candidates identify connectivity, stereochemistry, salt/solvate and hydration state, and purity. Distinct connectivity is separately indexed even if elemental fractions match.

[0036] For polymers specify repeat units, incorporated comonomer ratios, sequence distribution, tacticity, molecular-weight distribution, branching, end groups, crosslink topology, additives, and conditioning. Feed composition alone does not establish incorporated composition. For composites specify constituent identities, dimensions, orientation, interfaces, dispersion, layer order, void fraction, and the basis of each fraction.

[0037] Spatial variations can be represented by a composition field $x_i(r,t)$ with a defined coordinate region and sampling or averaging convention. A whole-specimen fraction is an integral over the appropriate count or mass density, not a simple average of unequal regions. Gradient, core-shell, layered, patterned, and porous descriptions require their dimensions and fabrication information. None is assumed to arise from every overall vector.

8 Preparation and confirmation records

[0038] Each preparation record states starting substances, grades, quantities, apparatus, order of operations, temperature, pressure, atmosphere, time, cooling or curing history, isolation, purification, post-treatment, scale, and relevant storage. Distinguish actual parameters from suggested values. A route class such as melting or deposition is insufficient as the preparation instruction for an unfamiliar composition.

[0039] Metals and alloys may be investigated through documented melting, powder consolidation, electrodeposition, or vapor deposition, followed by specified thermal and mechanical treatment. The selected route must address volatilization, immiscibility, reaction, contamination, competing phases, and phase stability for the actual constituents. This is a route-selection framework, not an assertion that one process works for every support.

[0040] Inorganic and ceramic candidates may use documented solid-state, solution, hydrothermal, solvothermal, deposition, or precursor-conversion routes. The actual precursors, stoichiometry, atmosphere, thermal schedule, isolation and phase-confirmation steps are necessary. Molecular candidates require a defined reaction or assembly and purification. Polymers require a defined polymerization, incorporated composition and forming history. Composites require constituent preparation, mixing or assembly, consolidation, and interface controls.

[0041] An unperformed recipe is a PROPOSED example stated prospectively. It reports no fabricated yield, measurement, or success. If a step depends on discovering an unavailable precursor, synthesizing a hypothetical element, stabilizing an unobserved nucleus, or establishing an unknown reaction, that dependency is an unresolved research problem rather than routine implementation.

[0042] Confirmation is selected to distinguish the target from realistic alternatives. Composition analysis alone may not identify a polymorph; diffraction alone may not detect a trace constituent; nominal instrument resolution is not a validated limit of detection in the actual matrix. Record calibration, reference materials, uncertainty, sampling, specimen history, and the method's detection and quantification limits. A nondetection is not proof of absence. [IUPAC limit of detection](#)

[0043] A tiny exact fraction is a mathematical target until preparation and measurement support a physical assertion. Experimental tolerance is specified separately, for example an interval or uncertainty with confidence convention. No instrument is claimed here to distinguish compositions separated by 10^{-19} fraction. At trace levels, isotope signatures, contamination control, adsorption, sampling statistics, and atom-count discreteness can dominate the interpretation.

[0044] An established teaching for comparison is TiO₂(B) nanosheets reported by Xiang and colleagues in 2013. Their nominal formula has Ti:O = 1:2; the B phase, nanosheet morphology, and exposed surface are additional features. Their paper reports a solvothermal preparation from titanium tetrachloride, water, and ethylene glycol, at 150 degrees C for six hours, with isolation and washing and diffraction/microscopy characterization. This is an attributed source summary; the complete methods remain necessary, and no applicant repetition is claimed. [Xiang et al., Scientific Reports 3, 1411](#)

[0045] The TiO₂ formula corresponds to rational atomic fractions (1/3,2/3). Its inclusion demonstrates why the unrestricted rational domain is needed and why a decimal grid alone is incomplete. It does not establish exact product purity, all titanium oxide polymorphs, every Ti/O ratio, or every process and use. Reproducing old literature is not represented as creating a newly available teaching.

9 Isotopes and nuclear states

[0046] An isotope-resolved constituent is identified by atomic number Z, mass number A, neutron count N = A-Z, and nuclear-state designation. For the elemental vocabulary, Z is 1 through 118 or an explicitly hypothetical 119 through 138. A is an integer at least Z. This parameterization includes candidate identifiers beyond observed nuclides; it does not assert that every (Z,A) nucleus is bound, observable, stable, or preparable.

[0047] For each element i, let $y_{i,a,s}$ be the conditional fraction of its atoms assigned to isotope a and nuclear state s, with $y \geq 0$ and $\sum(y_{i,a,s}) = 1$ over the stated isotope inventory. The whole-sample atomic fraction of that nuclide-state is $x_i y_{i,a,s}$. Natural, enriched, depleted, pure-isotope target, and mixed-isotope distributions can be described. Availability, lifetime, decay products, state preparation, and time since preparation are separate requirements.

[0048] Ground states, isomers, resonances, and evaluated excited levels are not collapsed into one identifier. NUBASE2020 and AME2020 provide dated evaluation resources for nuclear states and atomic masses. Their measured, estimated, and unobserved entries must retain those distinctions. The 2020 edition's experimental cutoff is not a statement that it contains every nuclide known in September 2026. [NUBASE2020](#), [AME2020 Part I](#), [AME2020 Part II](#)

[0049] An archival isotope record retains the source edition, nuclide identifier, state, existence-status evidence, property-estimation markers, units, uncertainties, half-life, decay modes, and retrieval date. A systematics marker on a mass is not by itself an existence-status classification. LiveChart API records likewise require their evaluation provenance; its guide identifies the underlying snapshots. No complete current nuclide export is represented as included in this draft. [IAEA LiveChart API guide](#)

[0050] A changing radioactive inventory is a function of time and preparation history. Its composition vector is identified at a specified time or over a stated averaging period. Daughter products enter the support when

included in the denominator. A short-lived nuclear species is not assumed to persist through an ordinary bulk preparation or measurement.

10 Subatomic and emergent particle systems

[0051] Established elementary-particle categories are distinguished from composite particles, nuclei, material excitations, and hypothetical particles. The Standard Model register includes six quark flavors, six lepton flavors, gauge bosons, and the Higgs boson, with antiparticle and charge-state conventions where applicable. A category listing does not teach arbitrary stable macroscopic mixtures of those particles. [CERN Standard Model](#)

[0052] A particle-system record identifies species and experimental status, quantum numbers, charge, mass where established, energy/momentum distribution, population or occupation convention, confinement or host, temperature or non-equilibrium state, lifetime, interaction model, preparation, and observation. Quarks within a nucleus, that nucleus, and the atom containing it are not independently added to an elemental count unless a physical construction and separate accounting basis are defined.

[0053] Composite particles such as protons, neutrons, mesons, and nuclei require their own identities and preparation or host descriptions. Photons and other field excitations require frequency or energy and occupation conditions rather than an unqualified chemical mass fraction. Antimatter-containing candidates require containment and annihilation considerations; no ordinary stable alloy is inferred from a particle list.

[0054] Phonons, magnons, excitons, polarons, and related quasiparticles are host-dependent excitations. Their record specifies the underlying material, excitation mechanism, field or carrier conditions, lifetime, and observable. They do not extend the periodic table and are not interchangeable elemental feedstocks. [IUPAC quasiparticle definition](#)

[0055] Hypothetical classes include gravitons, axion-like particles, sterile-neutrino candidates, dark photons, magnetic-monopole candidates, supersymmetric partners, and other model-defined entities. Each is an UNSUPPORTED candidate unless evidence and a preparation/observation route are supplied. A model identifier, parameters, conservation constraints, and search status must accompany it. No finite list can establish all future theoretical particle species or enable their material combinations.

[0056] PDG2026 tables and the versioned PDG API are source locators for particle identities and properties. They include experimental searches and limits as well as established particles. Inclusion in a table is not proof of discovery. The cited 2026 listings have a stated cutoff, and updates need their own date. [PDG 2026 tables](#), [PDG API](#)

11 Conditional properties and alternative responses

[0057] A property scenario identifies a baseline material and state, a controlled change, the expected response with mechanism and conditions, a credible alternative response with its mechanism, a discriminating test, and evidence status. The baseline and alternative are separate conditional embodiments or hypotheses. They are not asserted simultaneously for the same specimen under identical conditions.

[0058] Mechanical scenarios include stiffness, shear modulus, yield and ultimate strength, torsional response, fracture toughness, ductility, hardness, fatigue, creep, wear, and stress relaxation. These quantities have distinct definitions. A proposed strength increase cannot be inferred from stiffness alone, and specimen geometry can affect torque resistance. Tests specify direction, temperature, strain rate, environment, conditioning, and failure criterion.

[0059] Electrical scenarios include DC and AC conductivity, resistivity, carrier type and density, mobility, dielectric response, capacitance, breakdown, piezoresponse, thermoelectric response, and superconducting behavior where supported. Frequency, field, temperature, contacts, and geometry are specified. Negative differential resistance under driven conditions is not treated as negative passive DC conductivity. No assertion of superconductivity or a transition temperature is made for an unidentified composition.

[0060] Additional scenario fields include thermal conductivity and expansion; heat capacity and transitions; optical absorption, emission, refractive and nonlinear response; magnetic ordering, susceptibility and coercivity; corrosion, oxidation and chemical stability; catalytic selectivity and rate; diffusion, sorption, permeability and wetting; radiation response; and biological interaction where relevant. Each requires a defined observable, conditions, model or measurement, and local evidence label. A category name is not a predicted performance value.

[0061] A scenario may propose increased, decreased, approximately unchanged, threshold, nonmonotonic, hysteretic, anisotropic, transient, or sign-changing response, only with an explicit baseline and physically meaningful definition. An unchanged result means within stated resolution or tolerance, not mathematically identical under every test. These response categories enumerate research questions; they do not establish all future unexpected results.

[0062] ESTABLISHED mechanism example with PROPOSED extrapolation: in suitable precipitation-strengthened alloys, fine precipitates can impede dislocations; altered aging can coarsen precipitates and reduce hardness. For a new alloy record, specify its phases and aging path, then test hardness and microscopy rather than asserting every added element strengthens it. [NBS study of 2024 aluminum alloy processing](#)

[0063] ESTABLISHED mechanism example with PROPOSED extrapolation: a cooling path in a glass-forming system may suppress crystallization, while a different path or geometry permits nucleation and growth. A candidate record specifies the melt and measured thermal history and distinguishes glass from crystalline product using diffraction and calorimetry. This does not identify glass-forming behavior for every support. [Johnson et al., metallic-glass formation](#)

[0064] ESTABLISHED mechanism example with PROPOSED extrapolation: particular defects in TiO₂ can affect carrier separation and photocatalytic pathways, while trapping and recombination can lower useful activity. A new proposal must identify the defect location and material state and separately measure carrier dynamics and reaction rate. Increased absorption is not automatically increased photocatalysis. [Defects and photocatalysis in rutile TiO₂](#)

[0065] ESTABLISHED mechanism example with PROPOSED extrapolation: effective polymer crosslinks can increase stiffness, but nominal added crosslinker does not determine the number of active crosslinks or network architecture. A proposed opposing response is tested through conversion, swelling, and modulus at controlled temperature and frequency. [Crosslinking of fibrous hydrogels](#)

[0066] ESTABLISHED mechanism example with PROPOSED extrapolation: conductive fillers can form transport networks; dispersion, contact barriers, and competing network mechanisms can change conductivity versus loading. The same named constituents do not fix the percolation threshold. A proposal records morphology and contact conditions alongside electrical measurements. [Competing electrical-percolation behavior](#)

[0067] ESTABLISHED mechanism example with PROPOSED extrapolation: aggregation can increase emission for suitable luminogens by changing nonradiative pathways, while other packing and excited-state interactions can quench it. A new candidate needs molecular identity, packing evidence, quantum yield, lifetime, and conditions. The alternatives do not apply indiscriminately to every compound. [Tetrathienylethene emission study](#)

[0068] For an otherwise identical material, an unrecognized property may already be inherent, but necessity must be established for the relevant conditions. Stating a property and its opposite as possible does not establish inherency. A later process or use with additional steps is separately compared. The property-scenario framework is not a promise to prevent patents on every later finding. [USPTO inherency guidance](#)

12 Worked numerical embodiments

Binary percent cells

[0069] CALCULATED example NUM-001 selects an arbitrary ordered pair of distinct constituent labels A and B, on an atomic basis. For every $j = 0, \dots, 99$, x_A is in $[j/100, (j+1)/100]$ and $x_B = 1 - x_A$. Every composition satisfying $0 < x_A < 1$ has two-element support, including shared internal cell endpoints. Only $x_A = 0$ or $x_A = 1$ is unary. No particular chemical preparation is inferred. The pair can be replaced by any ordered distinct pair from the specified vocabulary, with hypothetical labels retaining their status.

Fine trace range

[0070] CALCULATED example NUM-002 selects $x_A = 10/D, 11/D, \dots, 100/D$ and $x_B = 1 - x_A$, with $D = 10^{19}$. For instance $x_A = 10/D = 10^{-18}$ fraction = $10^{-16}\%$, and $x_B = (D-10)/D$. Exact fractions are stored as integer numerator and denominator strings, avoiding floating-point rounding of the complement. Every contiguous subrange $[l/D, u/D]$ with $10 \leq l \leq u \leq 100$ is separately addressable. Preparation and detectability remain UNSUPPORTED.

Rational composition outside the decimal grid

[0071] CALCULATED example NUM-003 uses $(1/3, 2/3)$. No integer n satisfies $n/10^{19} = 1/3$ because 3 does not divide 10^{19} . The example is explicitly included through unrestricted rational denominators. A molecular stoichiometry, phase, and product purity require additional descriptions; the vector alone is not the $\text{TiO}_2(\text{B})$ literature product.

Multiple components and rejected intervals

[0072] CALCULATED example NUM-004 uses the feasible ternary box in paragraph [0032]. NUM-005 rejects three lower bounds of 40% because their sum is 120%. NUM-006 selects any k from 2 through 138, assigns $1/D$ to $k-1$ constituents and $(D-k+1)/D$ to the last. For $k = 138$ the small constituents total $137/D$. An equal-fraction candidate uses $x_i = 1/k$ in the rational domain; it lies on the D grid only when k divides D .

13 Reproducible enumeration instructions

[0073] To enumerate supports, fix the vocabulary edition, choose k , and list every increasing k -tuple of atomic numbers. Attach each label's recognized or hypothetical status. To enumerate a fixed rational denominator q , list every positive integer k -tuple summing to q . Use the last numerator as q minus the sum of the preceding numerators and reject values below one. This procedure terminates for each finite q and k .

[0074] To enumerate the unrestricted rational domain, repeat for $q = k, k+1, \dots$ and retain only primitive representations when deduplication is required. This is a countably infinite sequence, not a claim that a finite application prints every tuple. To enumerate decimal intervals, iterate p , l , and u over the finite limits in paragraph [0023], form component boxes, apply normalization feasibility, and construct a witness. The real continuous domain is defined symbolically; it cannot be exhaustively listed by a finite digital enumeration.

[0075] Every generated record includes vocabulary edition, support tuple, composition basis, exact endpoints, normalization rule, witness or infeasibility reason, and evidence status. A record identifier encodes these

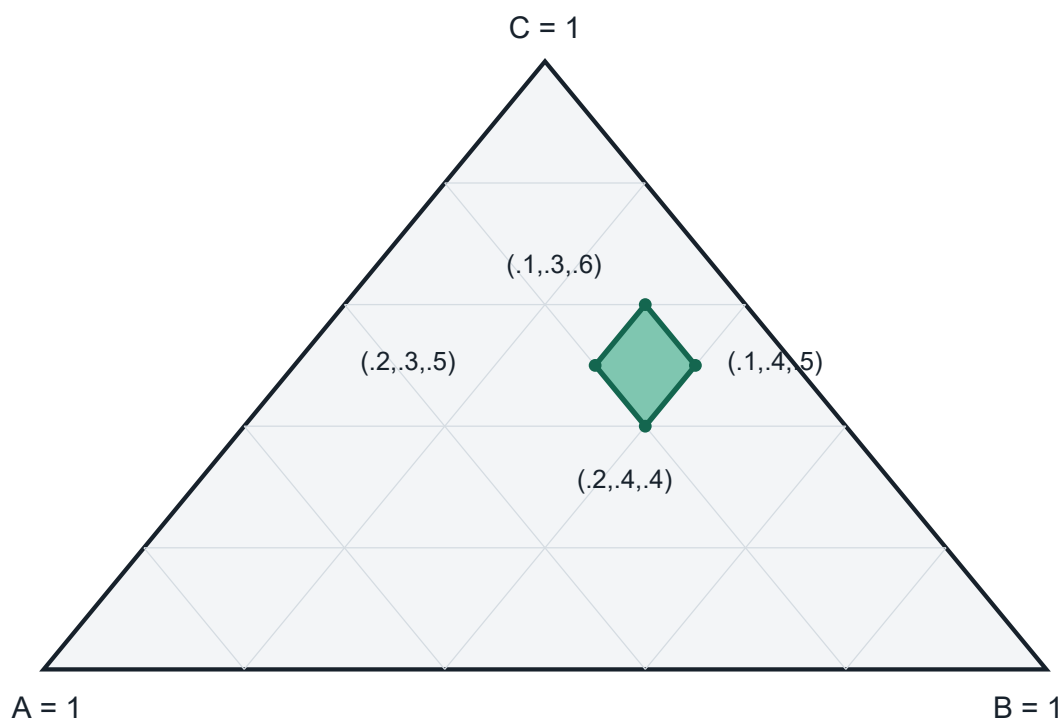
inputs, not an assertion of a synthesized material. Material identity, structure, preparation, and property fields are added independently and must not default to successful or established.

[0076] A reproducibility package accompanies this draft for exact arithmetic and verification. Its external files are aids for review, not automatically part of a filed or published application. Essential definitions and worked examples are therefore stated in this description. A final filing must carry all relied-on technical content in accepted application formats, with the filed and published versions checked.

14 Drawings

[0077] Figure 1 illustrates normalization for three constituents. The triangle represents the nonnegative simplex; its vertices are unary boundaries. The highlighted feasible interval intersection represents A in [10%,20%], B in [30%,40%], and C in [40%,60%]. It is a numerical diagram, not a measured phase diagram.

Figure 1. Normalized three-component composition

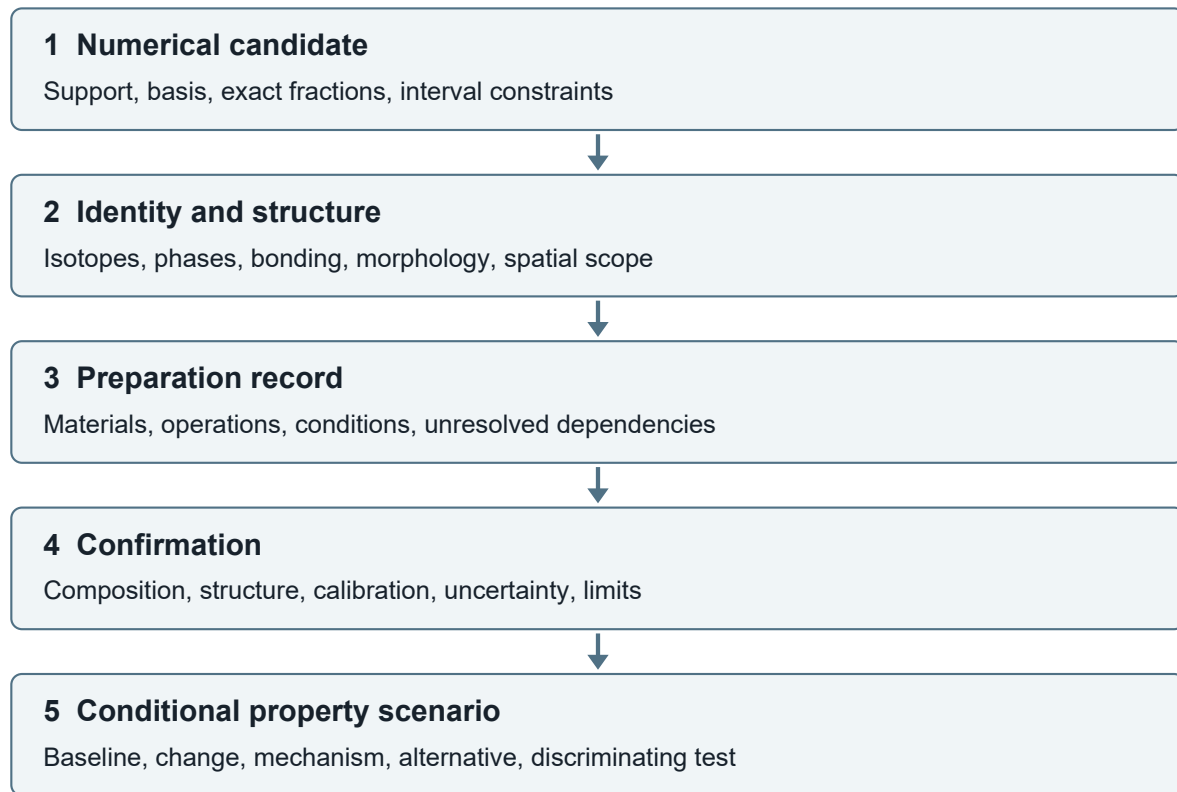


Highlighted: A in [0.1,0.2], B in [0.3,0.4], C in [0.4,0.6].

Every point obeys $A + B + C = 1$. Numerical region only; not a phase diagram.

[0078] Figure 2 shows the material-record sequence from a numerical candidate to structure, preparation, confirmation, and a conditional property scenario. An unresolved field remains unresolved. The drawing does not assert that this workflow supplies a universal preparation route.

Figure 2. Candidate-material record sequence



At every step: preserve evidence status and leave missing information unresolved.

A completed numerical record is not proof of an available physical material.

15 Industrial applicability and unresolved technical scope

[0079] Supported individual materials may be applicable to structural components, electronic and optical devices, thermal management, chemical processing, coatings, storage, or other uses established for the individual record. This draft does not establish industrial applicability of every hypothetical element, nuclide, particle system, or composition. An actual material-use record must state how the identified product is made and used.

[0080] Unresolved technical scope includes preparation of most indexed compositions; crystal and molecular identity; phase stability and coexistence; isotope availability and lifetime; precision and impurity control; particle confinement; property magnitudes and conditions; and the applicant's actual technical contribution. Numeric refinement cannot resolve these gaps. The candidate claims below express the requested scope for review and are not certified as novel, enabled, unified, or allowable.

16 Draft abstract

A framework describes candidate materials through elemental support sets, normalized composition vectors, interval constraints, isotope distributions, structures, preparation records, and conditional property scenarios. The vocabulary separates 118 recognized elements from hypothetical atomic-number placeholders through 138. Composition descriptions include positive continuous targets, unrestricted rational proportions, a decimal count grid with denominator 10^{19} , and correlated component ranges intersected with normalization. Exact endpoints, binary complements, and feasibility witnesses support reproducible indexing. Nuclear states and particle systems use separate identity and accounting conventions. Material records associate numerical candidates with preparation, characterization, evidence status, and unresolved conditions. The framework distinguishes calculated descriptions and unperformed proposals from established material teachings.

17 Candidate claims

The following complete claim annex expresses requested coverage for review. Hypothetical constituents and broad material classes lack demonstrated full-scope preparation support. Adding separate count or interval claims does not cure that deficiency.

Novelty has not been assessed for any of claims 1-199. Claims 1-185 are physical-material review candidates; claims 186-193 and 197-199 concern record operations, and claims 194-196 concern separate particle-system records. No claim represents an assertion that its full physical scope is enabled.

Claim 1. A material comprising a closed elemental inventory of k distinct elemental constituents selected from atomic numbers 1 through 138, wherein k is an integer from 2 through 138, each selected constituent has a positive normalized fraction x_i on an expressly identified atomic-fraction or mass-fraction basis, the fractions of the selected constituents sum to one, and constituents outside the selected inventory have zero fraction on that basis.

Claim 2. The material of claim 1, wherein k is exactly 2.

Claim 3. The material of claim 1, wherein k is exactly 3.

Claim 4. The material of claim 1, wherein k is exactly 4.

Claim 5. The material of claim 1, wherein k is exactly 5.

Claim 6. The material of claim 1, wherein k is exactly 6.

Claim 7. The material of claim 1, wherein k is exactly 7.

Claim 8. The material of claim 1, wherein k is exactly 8.

Claim 9. The material of claim 1, wherein k is exactly 9.

Claim 10. The material of claim 1, wherein k is exactly 10.

Claim 11. The material of claim 1, wherein k is exactly 11.

Claim 12. The material of claim 1, wherein k is exactly 12.

Claim 13. The material of claim 1, wherein k is exactly 13.

Claim 14. The material of claim 1, wherein k is exactly 14.

Claim 15. The material of claim 1, wherein k is exactly 15.

Claim 16. The material of claim 1, wherein k is exactly 16.

Claim 17. The material of claim 1, wherein k is exactly 17.

Claim 18. The material of claim 1, wherein k is exactly 18.

Claim 19. The material of claim 1, wherein k is exactly 19.

Claim 20. The material of claim 1, wherein k is exactly 20.

Claim 21. The material of claim 1, wherein k is exactly 21.

Claim 22. The material of claim 1, wherein k is exactly 22.

Claim 23. The material of claim 1, wherein k is exactly 23.

Claim 24. The material of claim 1, wherein k is exactly 24.

Claim 25. The material of claim 1, wherein k is exactly 25.

Claim 26. The material of claim 1, wherein k is exactly 26.

Claim 27. The material of claim 1, wherein k is exactly 27.

Claim 28. The material of claim 1, wherein k is exactly 28.

Claim 29. The material of claim 1, wherein k is exactly 29.

Claim 30. The material of claim 1, wherein k is exactly 30.

Claim 31. The material of claim 1, wherein k is exactly 31.

Claim 32. The material of claim 1, wherein k is exactly 32.

Claim 33. The material of claim 1, wherein k is exactly 33.

Claim 34. The material of claim 1, wherein k is exactly 34.

Claim 35. The material of claim 1, wherein k is exactly 35.

Claim 36. The material of claim 1, wherein k is exactly 36.

Claim 37. The material of claim 1, wherein k is exactly 37.

Claim 38. The material of claim 1, wherein k is exactly 38.

Claim 39. The material of claim 1, wherein k is exactly 39.

Claim 40. The material of claim 1, wherein k is exactly 40.

Claim 41. The material of claim 1, wherein k is exactly 41.

Claim 42. The material of claim 1, wherein k is exactly 42.

Claim 43. The material of claim 1, wherein k is exactly 43.

Claim 44. The material of claim 1, wherein k is exactly 44.

Claim 45. The material of claim 1, wherein k is exactly 45.

Claim 46. The material of claim 1, wherein k is exactly 46.

Claim 47. The material of claim 1, wherein k is exactly 47.

Claim 48. The material of claim 1, wherein k is exactly 48.

Claim 49. The material of claim 1, wherein k is exactly 49.

Claim 50. The material of claim 1, wherein k is exactly 50.

Claim 51. The material of claim 1, wherein k is exactly 51.

Claim 52. The material of claim 1, wherein k is exactly 52.

Claim 53. The material of claim 1, wherein k is exactly 53.

Claim 54. The material of claim 1, wherein k is exactly 54.

Claim 55. The material of claim 1, wherein k is exactly 55.

Claim 56. The material of claim 1, wherein k is exactly 56.

Claim 57. The material of claim 1, wherein k is exactly 57.

Claim 58. The material of claim 1, wherein k is exactly 58.

Claim 59. The material of claim 1, wherein k is exactly 59.

Claim 60. The material of claim 1, wherein k is exactly 60.

Claim 61. The material of claim 1, wherein k is exactly 61.

Claim 62. The material of claim 1, wherein k is exactly 62.

Claim 63. The material of claim 1, wherein k is exactly 63.

Claim 64. The material of claim 1, wherein k is exactly 64.

Claim 65. The material of claim 1, wherein k is exactly 65.

Claim 66. The material of claim 1, wherein k is exactly 66.

Claim 67. The material of claim 1, wherein k is exactly 67.

Claim 68. The material of claim 1, wherein k is exactly 68.

Claim 69. The material of claim 1, wherein k is exactly 69.

Claim 70. The material of claim 1, wherein k is exactly 70.

Claim 71. The material of claim 1, wherein k is exactly 71.

Claim 72. The material of claim 1, wherein k is exactly 72.

Claim 73. The material of claim 1, wherein k is exactly 73.

Claim 74. The material of claim 1, wherein k is exactly 74.

Claim 75. The material of claim 1, wherein k is exactly 75.

Claim 76. The material of claim 1, wherein k is exactly 76.

Claim 77. The material of claim 1, wherein k is exactly 77.

Claim 78. The material of claim 1, wherein k is exactly 78.

Claim 79. The material of claim 1, wherein k is exactly 79.

Claim 80. The material of claim 1, wherein k is exactly 80.

Claim 81. The material of claim 1, wherein k is exactly 81.

Claim 82. The material of claim 1, wherein k is exactly 82.

Claim 83. The material of claim 1, wherein k is exactly 83.

Claim 84. The material of claim 1, wherein k is exactly 84.

Claim 85. The material of claim 1, wherein k is exactly 85.

Claim 86. The material of claim 1, wherein k is exactly 86.

Claim 87. The material of claim 1, wherein k is exactly 87.

Claim 88. The material of claim 1, wherein k is exactly 88.

Claim 89. The material of claim 1, wherein k is exactly 89.

Claim 90. The material of claim 1, wherein k is exactly 90.

Claim 91. The material of claim 1, wherein k is exactly 91.

Claim 92. The material of claim 1, wherein k is exactly 92.

Claim 93. The material of claim 1, wherein k is exactly 93.

Claim 94. The material of claim 1, wherein k is exactly 94.

Claim 95. The material of claim 1, wherein k is exactly 95.

Claim 96. The material of claim 1, wherein k is exactly 96.

Claim 97. The material of claim 1, wherein k is exactly 97.

Claim 98. The material of claim 1, wherein k is exactly 98.

Claim 99. The material of claim 1, wherein k is exactly 99.

Claim 100. The material of claim 1, wherein k is exactly 100.

Claim 101. The material of claim 1, wherein k is exactly 101.

Claim 102. The material of claim 1, wherein k is exactly 102.

Claim 103. The material of claim 1, wherein k is exactly 103.

Claim 104. The material of claim 1, wherein k is exactly 104.

Claim 105. The material of claim 1, wherein k is exactly 105.

Claim 106. The material of claim 1, wherein k is exactly 106.

Claim 107. The material of claim 1, wherein k is exactly 107.

Claim 108. The material of claim 1, wherein k is exactly 108.

Claim 109. The material of claim 1, wherein k is exactly 109.

Claim 110. The material of claim 1, wherein k is exactly 110.

Claim 111. The material of claim 1, wherein k is exactly 111.

Claim 112. The material of claim 1, wherein k is exactly 112.

Claim 113. The material of claim 1, wherein k is exactly 113.

Claim 114. The material of claim 1, wherein k is exactly 114.

Claim 115. The material of claim 1, wherein k is exactly 115.

Claim 116. The material of claim 1, wherein k is exactly 116.

Claim 117. The material of claim 1, wherein k is exactly 117.

Claim 118. The material of claim 1, wherein k is exactly 118.

Claim 119. The material of claim 1, wherein k is exactly 119.

Claim 120. The material of claim 1, wherein k is exactly 120.

Claim 121. The material of claim 1, wherein k is exactly 121.

Claim 122. The material of claim 1, wherein k is exactly 122.

Claim 123. The material of claim 1, wherein k is exactly 123.

- Claim 124.** The material of claim 1, wherein k is exactly 124.
- Claim 125.** The material of claim 1, wherein k is exactly 125.
- Claim 126.** The material of claim 1, wherein k is exactly 126.
- Claim 127.** The material of claim 1, wherein k is exactly 127.
- Claim 128.** The material of claim 1, wherein k is exactly 128.
- Claim 129.** The material of claim 1, wherein k is exactly 129.
- Claim 130.** The material of claim 1, wherein k is exactly 130.
- Claim 131.** The material of claim 1, wherein k is exactly 131.
- Claim 132.** The material of claim 1, wherein k is exactly 132.
- Claim 133.** The material of claim 1, wherein k is exactly 133.
- Claim 134.** The material of claim 1, wherein k is exactly 134.
- Claim 135.** The material of claim 1, wherein k is exactly 135.
- Claim 136.** The material of claim 1, wherein k is exactly 136.
- Claim 137.** The material of claim 1, wherein k is exactly 137.
- Claim 138.** The material of claim 1, wherein k is exactly 138.
- Claim 139.** A single-element material having a closed elemental inventory consisting of one constituent selected from atomic numbers 1 through 138, wherein the constituent has normalized atomic fraction one and every other elemental constituent has fraction zero.
- Claim 140.** The material of claim 1, wherein every selected constituent has atomic number from 1 through 118.
- Claim 141.** The material of claim 1, wherein at least one selected constituent is selected from atomic numbers 119 through 138.
- Claim 142.** The material of claim 1, wherein k is 118 and the inventory includes each atomic number from 1 through 118.
- Claim 143.** The material of claim 1, wherein k is 138 and the inventory includes each atomic number from 1 through 138.
- Claim 144.** The material of claim 1, wherein each x_i is the number of atoms of constituent i divided by the total number of atoms in the specified inventory.
- Claim 145.** The material of claim 1, wherein the proportions are specified as mole-of-atoms fractions using the same atom inventory as an atomic-fraction description.
- Claim 146.** The material of claim 1, wherein each x_i is the mass of constituent i divided by the total mass of the specified inventory.
- Claim 147.** The material of claim 1, wherein an isotope-qualified mass M_i is specified for every constituent and the atomic and mass fractions are related by $w_i = x_i M_i / \sum(x_j M_j)$.
- Claim 148.** The material of claim 1, wherein $x_i = n_i/q$ for a positive integer q and positive integers n_i whose sum is q, with q unrestricted except that q is at least k.
- Claim 149.** The material of claim 148, wherein $\gcd(q, n_1, \dots, n_k) = 1$.

Claim 150. The material of claim 1, wherein $x_i = n_i/D$ with $D = 10^{19}$, each n_i is a positive integer, and $\sum(n_i) = D$.

Claim 151. The material of claim 150, wherein $k-1$ constituents each have fraction $1/D$ and the remaining constituent has fraction $(D-k+1)/D$.

Claim 152. The material of claim 1, wherein each fraction satisfies a selected closed interval $[L_i, U_i]$, with $0 \leq L_i \leq U_i \leq 1$, and the joint composition satisfies $\sum(x_i) = 1$ and $x_i > 0$.

Claim 153. The material of claim 152, wherein each interval is selected by integers p, l, u with $0 \leq p \leq 19$ and $0 \leq l \leq u \leq 10^p$, its endpoints being $l/10^p$ and $u/10^p$.

Claim 154. The material of claim 153, wherein at least one interval is an elementary cell $[j/10^p, (j+1)/10^p]$ for an integer j with $0 \leq j < 10^p$.

Claim 155. The material of claim 1, wherein $k = 2$ and fractions x_A and x_B obey x_A in $[L, U]$ and $x_B = 1 - x_A$ in $[1 - U, 1 - L]$, with $0 < x_A < 1$.

Claim 156. The material of claim 155, wherein $10^{-18} \leq x_A \leq 10^{-17}$, corresponding to 10⁻¹⁶% through 10⁻¹⁵%, inclusive.

Claim 157. The material of claim 156, wherein x_A is exactly 10^{-18} and x_B is exactly $1 - 10^{-18}$.

Claim 158. The material of claim 156, wherein x_A is exactly 10^{-17} and x_B is exactly $1 - 10^{-17}$.

Claim 159. The material of claim 156, wherein $x_A = n/10^{19}$ for an integer n from 10 through 100 and $x_B = 1 - x_A$.

Claim 160. The material of claim 159, wherein x_A lies in a selected subrange $[l/10^{19}, u/10^{19}]$ for integers $10 \leq l \leq u \leq 100$.

Claim 161. The material of claim 1, wherein $k = 2$ and each constituent has fraction $1/2$, corresponding to 50% each.

Claim 162. The material of claim 1, wherein $k = 2$ and the constituent fractions are exactly $1/3$ and $2/3$.

Claim 163. The material of claim 1, wherein every constituent has fraction $1/k$.

Claim 164. The material of claim 1, wherein every constituent has fraction at least 10^{-19} .

Claim 165. The material of claim 1, wherein at least one constituent has positive fraction less than 10^{-19} .

Claim 166. The material of claim 152, wherein at least one interval is $[j \text{ times } 10^{(e-r)}, (j+1) \text{ times } 10^{(e-r)}]$, with integer e from -19 through -1, nonnegative integer r , and integer j from 10^r through $10^{(r+1)} - 1$.

Claim 167. The material of claim 152, wherein at least one interval has arbitrary rational endpoints r and s satisfying $0 \leq r \leq s \leq 1$.

Claim 168. The material of claim 152, wherein the endpoints are specified as real target values and the selected fractions satisfy their interval constraints and exact normalization.

Claim 169. The material of claim 1, wherein each selected element has a conditional isotope-state distribution $y_{i,a,s}$ with nonnegative entries summing to one and whole-inventory nuclide-state fractions $x_i y_{i,a,s}$.

Claim 170. The material of claim 169, wherein at least one constituent is specified by a single mass number and nuclear state with conditional fraction one.

Claim 171. The material of claim 169, wherein at least one constituent has two or more specified isotope-state entries of positive conditional fraction.

Claim 172. The material of claim 169, wherein each specified nuclear state has a source-qualified identifier that distinguishes a ground state from any specified isomer or excited state.

Claim 173. The material of claim 169, wherein a radioactive inventory is identified at a specified time and includes daughter constituents that fall within its stated denominator.

Claim 174. The material of claim 1, wherein a crystalline constituent phase has a specified structural model, site occupations, substitution rules and charge-balance constraints.

Claim 175. The material of claim 1, wherein an amorphous constituent phase is identified by a stated preparation history and a specified characterization convention for detectable crystalline fraction.

Claim 176. The material of claim 1, wherein molecular identity includes specified connectivity, stereochemistry, salt or solvate state, and hydration state where present.

Claim 177. The material of claim 1, wherein the material is a polymer specified by repeat units, incorporated ratios, sequence distribution, molecular-weight distribution and network architecture.

Claim 178. The material of claim 1, wherein the material is a composite specified by constituent identities, dimensions, orientations, interfaces and dispersion.

Claim 179. The material of claim 1, wherein the material has a specified layer order and layer dimensions, with local composition distinguished from whole-specimen composition.

Claim 180. The material of claim 1, wherein the material has a specified spatial composition field $x_i(r,t)$ and a density-weighted inventory convention for deriving the whole-specimen fractions.

Claim 181. The material of claim 1, wherein the stated elemental inventory describes a specified single phase.

Claim 182. The material of claim 1, wherein the stated elemental inventory describes a specified multiphase assemblage and each phase is separately identified.

Claim 183. The material of claim 178, wherein constituent volume fractions and void fraction are specified separately from elemental atomic fractions under stated temperature, pressure, density and volume-assignment conventions.

Claim 184. The material of claim 1, wherein a nominal target fraction is accompanied by a separately specified measurement uncertainty or tolerance and the associated sampling convention.

Claim 185. The material of claim 1, wherein the product composition is specified separately from its feed proportions and from local measured composition.

Claim 186. A method of constructing a candidate-material record, comprising selecting a constituent support and composition basis; assigning positive normalized target fractions whose sum is one; recording structural, preparation, measurement and property fields; and recording an evidence status for each field without treating an unresolved field as a successful preparation.

Claim 187. The method of claim 186, comprising enumerating every increasing k-tuple of selected atomic numbers for a specified k from 1 through 138 and attaching a recognized or hypothetical status to each label.

Claim 188. The method of claim 186, comprising enumerating positive integer tuples with sum q for successive unrestricted positive denominators q, and retaining primitive common-denominator representations.

Claim 189. The method of claim 186, comprising enumerating positive integer tuples with sum $D = 10^{19}$ and storing their normalized fractions as exact integer numerator and denominator strings.

Claim 190. The method of claim 186, comprising selecting every endpoint pair l, u with $0 \leq l \leq u \leq 10^p$ for each p from 0 through 19, forming correlated component boxes, and rejecting boxes that fail the applicable normalization and positivity tests.

Claim 191. The method of claim 186, comprising testing $0 \leq L_i \leq U_i \leq 1$ and $\sum(L_i) \leq 1 \leq \sum(U_i)$, then constructing a nonnegative witness by starting at the lower bounds and allocating residual mass without exceeding any upper bound.

Claim 192. The method of claim 191, further comprising checking every $U_i > 0$ for a strictly positive support; when $\sum(L_i) = 1$ requiring every $L_i > 0$; and when $\sum(L_i) < 1$ raising the lower bounds to $\max(L_i, \epsilon)$, where $\epsilon = \min(\min_i U_i, (1 - \sum(L_i))/(2k))$, before residual allocation.

Claim 193. The method of claim 186, comprising calculating $b_i = \max(1, \text{ceiling}(D L_i))$ and $c_i = \min(D - k + 1, \text{floor}(D U_i))$ for $D = 10^{19}$, accepting a grid box only when $b_i \leq c_i$ and $\sum(b_i) \leq D \leq \sum(c_i)$, and constructing an integer witness within those bounds.

Claim 194. A method of constructing a particle-system record, comprising identifying a species or model-qualified candidate, experimental status, quantum numbers, population or occupation convention, energy conditions, host or confinement, lifetime information, preparation and observation fields, while specifying a construction level that avoids counting a particle and its containing composite as independent copies of the same inventory.

Claim 195. The method of claim 194, wherein the record concerns a host-dependent excitation and specifies the host material, excitation mechanism, conditions, lifetime or linewidth, and observable.

Claim 196. The method of claim 194, wherein an unconfirmed entity is marked hypothetical, with a model identifier, parameter assumptions, conservation constraints and experimental-search status, without assigning unsupported measured properties.

Claim 197. The method of claim 186, comprising recording a property scenario with a baseline material and state, controlled change, proposed response, mechanism, conditions, discriminating test and evidence status.

Claim 198. The method of claim 197, comprising recording a credible alternative response and its mechanism as a separate conditional hypothesis, rather than asserting opposite responses for the same specimen under identical conditions.

Claim 199. The method of claim 186, comprising labeling attributed source teachings ESTABLISHED, stated mathematical or model results CALCULATED, unperformed tests or preparations PROPOSED, and possibilities without adequate support UNSUPPORTED, and retaining source and condition information with the labels.

18 Element and particle register

The following 138 rows are self-contained identity entries. Recognized rows follow the [IUPAC table dated 4 May 2022](#), checked 29 September 2026. E119-E138 are local hypothetical placeholders, not accepted symbols. No row supplies a making method or atomic mass.

Z	Symbol / label	Name / local description	Status
1	H	hydrogen	Recognized
2	He	helium	Recognized
3	Li	lithium	Recognized
4	Be	beryllium	Recognized
5	B	boron	Recognized
6	C	carbon	Recognized
7	N	nitrogen	Recognized
8	O	oxygen	Recognized
9	F	fluorine	Recognized
10	Ne	neon	Recognized
11	Na	sodium	Recognized
12	Mg	magnesium	Recognized
13	Al	aluminium	Recognized
14	Si	silicon	Recognized
15	P	phosphorus	Recognized
16	S	sulfur	Recognized
17	Cl	chlorine	Recognized
18	Ar	argon	Recognized
19	K	potassium	Recognized
20	Ca	calcium	Recognized
21	Sc	scandium	Recognized
22	Ti	titanium	Recognized
23	V	vanadium	Recognized
24	Cr	chromium	Recognized
25	Mn	manganese	Recognized
26	Fe	iron	Recognized

Z	Symbol / label	Name / local description	Status
27	Co	cobalt	Recognized
28	Ni	nickel	Recognized
29	Cu	copper	Recognized
30	Zn	zinc	Recognized
31	Ga	gallium	Recognized
32	Ge	germanium	Recognized
33	As	arsenic	Recognized
34	Se	selenium	Recognized
35	Br	bromine	Recognized
36	Kr	krypton	Recognized
37	Rb	rubidium	Recognized
38	Sr	strontium	Recognized
39	Y	yttrium	Recognized
40	Zr	zirconium	Recognized
41	Nb	niobium	Recognized
42	Mo	molybdenum	Recognized
43	Tc	technetium	Recognized
44	Ru	ruthenium	Recognized
45	Rh	rhodium	Recognized
46	Pd	palladium	Recognized
47	Ag	silver	Recognized
48	Cd	cadmium	Recognized
49	In	indium	Recognized
50	Sn	tin	Recognized
51	Sb	antimony	Recognized
52	Te	tellurium	Recognized
53	I	iodine	Recognized
54	Xe	xenon	Recognized
55	Cs	caesium	Recognized

Z	Symbol / label	Name / local description	Status
56	Ba	barium	Recognized
57	La	lanthanum	Recognized
58	Ce	cerium	Recognized
59	Pr	praseodymium	Recognized
60	Nd	neodymium	Recognized
61	Pm	promethium	Recognized
62	Sm	samarium	Recognized
63	Eu	europium	Recognized
64	Gd	gadolinium	Recognized
65	Tb	terbium	Recognized
66	Dy	dysprosium	Recognized
67	Ho	holmium	Recognized
68	Er	erbium	Recognized
69	Tm	thulium	Recognized
70	Yb	ytterbium	Recognized
71	Lu	lutetium	Recognized
72	Hf	hafnium	Recognized
73	Ta	tantalum	Recognized
74	W	tungsten	Recognized
75	Re	rhenium	Recognized
76	Os	osmium	Recognized
77	Ir	iridium	Recognized
78	Pt	platinum	Recognized
79	Au	gold	Recognized
80	Hg	mercury	Recognized
81	Tl	thallium	Recognized
82	Pb	lead	Recognized
83	Bi	bismuth	Recognized
84	Po	polonium	Recognized

Z	Symbol / label	Name / local description	Status
85	At	astatine	Recognized
86	Rn	radon	Recognized
87	Fr	francium	Recognized
88	Ra	radium	Recognized
89	Ac	actinium	Recognized
90	Th	thorium	Recognized
91	Pa	protactinium	Recognized
92	U	uranium	Recognized
93	Np	neptunium	Recognized
94	Pu	plutonium	Recognized
95	Am	americium	Recognized
96	Cm	curium	Recognized
97	Bk	berkelium	Recognized
98	Cf	californium	Recognized
99	Es	einsteinium	Recognized
100	Fm	fermium	Recognized
101	Md	mendelevium	Recognized
102	No	nobelium	Recognized
103	Lr	lawrencium	Recognized
104	Rf	rutherfordium	Recognized
105	Db	dubnium	Recognized
106	Sg	seaborgium	Recognized
107	Bh	bohrium	Recognized
108	Hs	hassium	Recognized
109	Mt	meitnerium	Recognized
110	Ds	darmstadtium	Recognized
111	Rg	roentgenium	Recognized
112	Cn	copernicium	Recognized
113	Nh	nihonium	Recognized

Z	Symbol / label	Name / local description	Status
114	Fl	flerovium	Recognized
115	Mc	moscovium	Recognized
116	Lv	livermorium	Recognized
117	Ts	tennessine	Recognized
118	Og	oganesson	Recognized
119	E119	hypothetical element 119	Hypothetical placeholder
120	E120	hypothetical element 120	Hypothetical placeholder
121	E121	hypothetical element 121	Hypothetical placeholder
122	E122	hypothetical element 122	Hypothetical placeholder
123	E123	hypothetical element 123	Hypothetical placeholder
124	E124	hypothetical element 124	Hypothetical placeholder
125	E125	hypothetical element 125	Hypothetical placeholder
126	E126	hypothetical element 126	Hypothetical placeholder
127	E127	hypothetical element 127	Hypothetical placeholder
128	E128	hypothetical element 128	Hypothetical placeholder
129	E129	hypothetical element 129	Hypothetical placeholder
130	E130	hypothetical element 130	Hypothetical placeholder
131	E131	hypothetical element 131	Hypothetical placeholder
132	E132	hypothetical element 132	Hypothetical placeholder
133	E133	hypothetical element 133	Hypothetical placeholder
134	E134	hypothetical element 134	Hypothetical placeholder
135	E135	hypothetical element 135	Hypothetical placeholder
136	E136	hypothetical element 136	Hypothetical placeholder
137	E137	hypothetical element 137	Hypothetical placeholder
138	E138	hypothetical element 138	Hypothetical placeholder

Particle category register

Six quark flavor categories, three charged-lepton particle/antiparticle categories, three neutrino flavor categories, and five boson categories. W⁺ and W⁻ are grouped; gluon color components are grouped. This is a table convention, not an absolute count of physical particles or quantum states.

Flavor labels differ from neutrino mass eigenstates. The register does not assert that the minimal massless-neutrino Standard Model explains observed neutrino masses, or resolve Dirac versus Majorana nature.

Sources: [CERN Standard Model](#) and [PDG 2026 tables](#). The 17 rows are categories under this convention, not all particle states or material ingredients.

Category	Labels	Accounting and status qualification
up quark flavor	u, anti-u	One flavor category; antiquark and three color states are not additional rows. No isolated free-quark ingredient is asserted.
down quark flavor	d, anti-d	One flavor category; antiquark and three color states are not additional rows. No isolated free-quark ingredient is asserted.
charm quark flavor	c, anti-c	One flavor category; antiquark and three color states are not additional rows. No isolated free-quark ingredient is asserted.
strange quark flavor	s, anti-s	One flavor category; antiquark and three color states are not additional rows. No isolated free-quark ingredient is asserted.
top quark flavor	t, anti-t	One flavor category; antiquark and three color states are not additional rows. No isolated free-quark ingredient is asserted.
bottom quark flavor	b, anti-b	One flavor category; antiquark and three color states are not additional rows. No isolated free-quark ingredient is asserted.
electron and positron	e-, e+	Particle and antiparticle grouped in one category.
muon and antimuon	mu-, mu+	Particle and antiparticle grouped in one category.
tau and antitau	tau-, tau+	Particle and antiparticle grouped in one category.
electron-flavor neutrino	nu_electron, anti-nu_electron	Weak-interaction flavor category, not a mass eigenstate. Labels for neutrino/antineutrino interactions do not settle Dirac versus Majorana nature.
muon-flavor neutrino	nu_muon, anti-nu_muon	Weak-interaction flavor category, not a mass eigenstate. Labels for neutrino/antineutrino interactions do not settle Dirac versus Majorana nature.
tau-flavor neutrino	nu_tau, anti-nu_tau	Weak-interaction flavor category, not a mass eigenstate. Labels for neutrino/antineutrino interactions do not settle Dirac versus Majorana nature.
photon	gamma	One species category; momentum and polarization states are not separately enumerated.
gluon	g	Eight color components grouped in one category; no free isolated-gluon ingredient is asserted.
charged W bosons	W+, W-	Two charge-conjugate states grouped in one category.
Z boson	Z0	One species category; polarization states are not separate entries.
Higgs boson	H	Observed Higgs-boson category; additional hypothetical Higgs states are not asserted.

Composite particles, nuclear states, quasiparticles and hypothetical particle classes use the separate parameters in paragraphs [0046]-[0056]. Their possible identifiers are not additional recognized elemental rows. No complete current nuclide table or exhaustive theoretical-particle universe is claimed.

19 Claim support and review map

Textual support and physical enablement are different inquiries. Every claim remains novelty-unassessed. The complete per-claim map is supplied in the companion claim_support_map.json; the operative definitions and claims are printed here and do not depend on that external file.

Claims	Textual locations	Subject matter	Enablement and review status
1-138	[0010]-[0022], [0073]	Broad normalized material inventory and every exact component count.	Mathematical definitions present; composition-specific full-scope preparation absent. Counts above 118 necessarily include hypothetical labels.
139-147	[0010]-[0016], [0041], [0047]	Unary material, vocabulary restrictions and composition bases.	Identity or basis definitions do not establish novelty, availability or making methods.
148-168	[0017]-[0032], [0069]-[0072]	Rational and grid fractions, all endpoint ranges, complements and explicit boundary targets.	Exact arithmetic present; physical realization, tolerances and phase identity unresolved. Below-grid targets are broader than the floor-limited verifier.
169-173	[0046]-[0050]	Isotope, nuclear-state and time-indexed inventory descriptions.	Parameter identifiers do not establish nuclear existence, isolation, lifetime or synthesis.
174-185	[0012]-[0013], [0033]-[0043]	Structures, phases, spatial forms and measurement conventions.	Structural fields are specified as requirements, not completed material teachings for the universal composition domain.
186-193	[0005]-[0007], [0023]-[0032], [0073]-[0075]	Candidate-record construction and mathematical enumeration/feasibility methods.	Operations are mathematically described; novelty, eligibility, inventive contribution and full legal support remain unassessed.
194-196	[0051]-[0056]	Separate particle-system record methods.	Recording method does not prepare or stabilize every particle or hypothetical system.
197-199	[0005], [0041]-[0042], [0057]-[0068], [0075]	Conditional property scenarios and local evidence status.	No measured applicant result, inherent property or universal predicted performance is supplied.

20 Source and version notes

Source links identify attributed teachings, definitions, and evaluation editions. Mathematical models, record designs, numerical examples, and proposed scenarios are this draft's calculations or proposals. Source retrieval and local document creation dates are not patent filing or public-availability dates. No applicant identity, inventor signature, priority claim, micro-entity certification, complete current isotope archive, or worldwide patent-status finding is assumed. The later filing memorandum addresses procedure separately from this technical description.

Technical scope supplement: compositions, finite inventories and dated entities

Prepared 30 September 2026 UTC. This supplement adds technical definitions, calculations and source-qualified records to the original application working draft (source file: provisional_application_draft.md). Its 80 numbered paragraphs, 199 candidate claims and original PDFs are retained. This supplement is not a filing receipt, a report of applicant experiments, or an assertion that the full physical domain is enabled.

The requested scope remains all selected elemental supports and proportions, all component counts through 138, isotope and particle qualifications, and conditional expected or contrary property responses. Numerical addressing, entity identity, physical preparation, observation and legal effect remain distinct layers. The full-scope audit (source file: full_scope_completion_audit.md) records what is present and what remains unproved.

Composition domains and exact addresses

The element vocabulary has recognized labels for $Z=1\dots 118$ and separate hypothetical placeholders for $Z=119\dots 138$. The [IUPAC periodic-table page](#) supplies the recognized table used here. The 138-label boundary is a chosen scope index; it is not a physical maximum or evidence that the last 20 nuclei exist. A support S is any nonempty strictly increasing tuple of distinct labels. A zero coordinate belongs to a smaller canonical support rather than an additional present constituent.

For a support containing k labels, distinguish the following domains:

Domain	Definition	Meaning
Strictly positive real target	$x_i > 0$ and $\sum x_i = 1$	A symbolic target domain, including irrational coordinates; it is not a finite list or an exact atom-count inventory.
Unrestricted rational target	$x_i = n_i/q$, n_i positive integers, $\sum n_i = q$, with q any positive integer at least k	Exact count-compatible candidate ratios with no imposed numerical concentration floor.
Primitive rational address	The rational definition with $\gcd(q, n_1, \dots, n_k) = 1$	A unique common-denominator address for a vector; prevents duplicated fractions from being treated as different compositions.
Optional decimal grid	$q = D = 10^{19}$ and each $n_i \geq 1$	The earlier requested finite grid. It remains useful but does not replace the unrestricted domain.
Measured composition	A stated assay, counted region, uncertainty and sampling convention	Experimental information, requiring actual evidence and a specified quantity basis.

The new unrestricted module (source file: ../range_model/unrestricted_compositions.py) implements the rational domain separately from the original bounded/grid module (source file: ../range_model/composition_ranges.py). Its exact APIs accept integer fractions and decimal/fraction strings, reject binary floats and booleans, and impose no 10^{19} floor. The original module retains that deliberate floor. Neither API numerically enumerates all real values.

For each fixed support, enumerate $q = k, k+1, \dots$; enumerate every positive integer tuple summing to q ; retain the primitive tuples. Every positive rational normalized vector appears exactly once. A command's maximum denominator or output limit bounds an export, not the defined domain. The procedure terminates for each finite q but does not claim that an infinite domain or the astronomical collection of all supports was printed in a finite application.

The finer examples include a binary fraction $1/10^{100}$ with its exact complement and equal proportions for all 138 labels. These are CALCULATED targets. No corresponding specimen, precursor availability or manufacturing precision is asserted. Counts above 118 necessarily use hypothetical labels.

All intervals and joint normalization

The original decimal hierarchy addresses every endpoint pair $[l/10^p, u/10^p]$ for $p=0\dots 19$, $0 \leq l \leq u \leq 10^p$, including singleton points, elementary cells and contiguous subranges. Arbitrary rational endpoints extend that hierarchy without a fixed numerical floor. Real endpoint targets remain symbolic. The interval selection must name its vocabulary, support and atomic, mass, volume, feed, site or other denominator.

At $p=2$, $[1\%, 2\%]$ in a binary material pairs with $[98\%, 99\%]$ for the other component. The two coordinates are correlated through $x_B = 1 - x_A$. The requested interval $0.0000000000000001\% \dots 0.0000000000000001\%$ equals fractions $10^{-18} \dots 10^{-17}$. On $D=10^{19}$ it contains numerators $10 \dots 100$, giving 91 grid points and 90 elementary bins. Percent and fractional values must retain their separate units. A fixed decimal denominator does not contain $1/3$; the unrestricted rational domain does.

For a closed rational component box, first intersect each interval with $[0, 1]$. A nonnegative normalized vector requires nonempty coordinate intervals and $\sum L_i \leq 1 \leq \sum U_i$. For a strictly positive support additionally require every $U_i > 0$. If $\sum L_i = 1$, every L_i must already be positive. If there is remaining slack and zero lower bounds, reserve a sufficiently small positive rational share for each zero lower bound, then allocate the residual within upper bounds. This constructs a positive witness without choosing a universal floor. A vector forced to include zero cannot inhabit that support.

For any selected positive integer denominator q , the exact count limits are $b_i = \max(1, \text{ceil}(q L_i))$ and $c_i = \text{floor}(q U_i)$. A lattice witness exists exactly when $b_i \leq c_i$ for every coordinate and $\sum b_i \leq q \leq \sum c_i$. Starting at the lower counts and allocating the residual constructs a witness. Continuous/rational feasibility does not imply feasibility on a particular q lattice. Failed interval requests are retained as rejected numerical candidates with their reason; they are not reported as materials.

The unrestricted verification report (source file: `../range_model/unrestricted_verification_report.json`) records 11 implementation check groups. Independent adversarial checks use box/simplex vertices, integer separator positions and explicit sample subsets rather than repeating the greedy algorithm. Mathematical proof and bounded tests establish their stated arithmetic scope. They do not establish preparation, phase stability or prior-art effect.

Finite atoms and joint isotope/state counts

For a specified N -atom inventory, each atomic fraction is N_i/N with integer $N_i \geq 0$ and $\sum N_i = N$. The smallest positive count fraction in that particular inventory is $1/N$. There is no specimen-independent lower fraction from this arithmetic. The [BIPM mole definition](#) supplies the exact Avogadro constant and requires the counted entity to be specified.

If the target vector has primitive denominator q , its exact counts in a fixed N -atom inventory exist at the accounting level exactly when q divides N . Counts are $N_i = N n_i/q$. Taking $N=q$ or a positive multiple resolves count compatibility; it does not resolve the physical existence, co-location, lifetime, controlled placement, analysis or utility of those atoms. Increasing N can permit a smaller nonzero fraction but does not establish an attainable experiment.

For element i , an isotope/state partition $y_{i,a,s}$ has nonnegative entries summing to one. A classical joint inventory additionally requires every $N \cdot x_i \cdot y_{i,a,s}$ to be an integer. Those sub-counts must sum to N_i . Element-level compatibility alone is insufficient. For example, elemental fractions $(1/3, 2/3)$ with anonymous conditional partitions $(1/2, 1/2)$ and $(1/4, 3/4)$ fit a six-atom inventory with sub-counts $(1, 1)$ and $(1, 3)$. They do not fit a three-atom inventory even though its elemental counts are integers.

conditional_inventory implements this calculation with exact fractions. Its partitions are anonymous accounting examples; they do not identify actual nuclides, prove a nuclear state exists, or establish a preparable quantum population. Zero conditional populations are allowed. Nuclear/particle labels, coherence or occupation conventions, and the validity of a classical mixture interpretation must be supplied separately. No unobserved nucleus inherits existence from an integer count.

Sampling is separate from detection

Suppose exactly M trace atoms exist among N atoms, and n distinct atoms are sampled uniformly without replacement, with integer $N \geq 1$, $0 \leq M \leq N$ and $0 \leq n \leq N$. Assuming the selection model, the probability none are included is $\text{binomial}(N-M, n) / \text{binomial}(N, n)$ when $n \leq N-M$, and zero otherwise. With one assigned trace atom the inclusion probability is exactly n/N . A one-percent subsample has a one-percent chance of containing it; 95% inclusion requires sampling at least 95% of that specimen. The module reports **sample inclusion**, leaving detector efficiency and classification unmodeled.

A different CALCULATED model takes independent draws from an unchanged effectively unlimited reservoir with trace probability p . Its count C is binomial: expectation np , variance $np(1-p)$, and probability zero $(1-p)^n$. For rare events with $np = \lambda$, the Poisson approximation gives $\exp(-\lambda)$. At $p = 10^{-19}$ and $n = 10^{19}$, the expected count is one, but the zero-count probability is approximately 0.3678794412. This independent-draw model does not describe repeated sampling of a fixed one-atom specimen without replacement.

For independent draws with $0 < p < 1$ and $0 < \alpha < 1$, at least one trace entity occurs with probability at least $1 - \alpha$ when $n \geq \lceil \log(\alpha) / \log(1-p) \rceil$. At $p = 0$ no finite number of draws gives a positive occurrence probability; at $p = 1$ one draw suffices. For $p = 10^{-19}$ and $\alpha = 0.05$, the high-precision calculation gives 29,957,322,735,539,909,933 draws. If zero counts are observed under perfect detection/classification in this model with $n \geq 1$, the exact one-sided 95% upper probability bound is $1 - 0.05^{(1/n)}$, approximately 2.995732274/ n at low concentrations. The calculation artifact (source file: `../range_model/trace_sampling_calculations.py`) states assumptions and verifies the adjacent minimum-count boundary.

For multinomial reservoir counts, $\text{Cov}(C_i, C_j) = -n p_i p_j$ for different classes. For uniform finite sampling without replacement, fixed-population variances and covariances acquire factor $(N-n)/(N-1)$ when $N > 1$. These are sampling models, not calibration results. Contamination, adsorption, detector efficiency, misclassification, background, correlated samples and selection bias require method-specific treatment. [NIST's metrological-traceability guidance](#) emphasizes uncertainty and fitness for purpose. A nondetection cannot prove an absolute zero constituent fraction merely because the target inventory is closed.

Physical integration constraints

The material record remains $R = (B, S, x, I, U, C, P, M, F, E)$, connecting composition basis, support, fractions, isotopes/states, structure, preparation, properties, measurements, failure boundaries and evidence. Numerical feasibility is one field. Add the following constraints independently rather than infer them from normalization:

Field	Required physical description	Inference not supported by a bare vector
Charge/construction level	Species charge, electronic state, boundary, countercharge/fields and a consistent counting level	Neutrality, confinement or stability of an arbitrary mixture; independent counting of an atom and its included nucleus/electrons.
Sites and stoichiometry	Site multiplicities, allowed occupants, vacancies, ordering and any compensation in a nominated phase	Occupancy of one lattice by every selected element or realizability of every bulk fraction in that phase.

Field	Required physical description	Inference not supported by a bare vector
Structure and geometry	Phases, connectivity, polymer sequence/network, interfaces, grain/particle dimensions, orientation and spatial distribution	A unique material identity from overall percentages.
History and persistence	Precursors, process conditions, scale, atmosphere, conditioning, competing reactions/phases, storage and observation duration	A universal melting, deposition or nuclear-production route; persistence through an unspecified measurement.
Measurement	Assay denominator, local versus whole-product sampling, calibration, uncertainty and detection/quantification limits	Exact manufacture or instrumental resolution at the numerical grid step.

For a region described only by nuclei and electrons, charge accounting is $Q/e = \sum Z N_Z - N_e$. A different particle construction uses $Q = \sum q_s N_s$ with signed species charges. Specify the construction level to avoid double counting. Neutrality is a constraint for a chosen neutral embodiment, not a universal prohibition on charged matter. Formal oxidation-state balancing is a stated chemical model, not a complete electronic-structure or stability calculation.

An atomic-to-mass conversion using isotope/state-qualified masses needs a declared allocation convention. Nominal neutral-atom masses, ion masses, independently weighed feeds and an interacting product are not automatically interchangeable at extreme precision. [AME2020 Part I](#), journal page 030002-7, discusses precision-dependent binding-energy corrections. This does not negate ordinary practical composition reporting; it prevents nominal numerical precision from being presented as exact physical metrology.

Dated nuclear and particle identity archives

The NUBASE2020 archive (source file: `../data/nuclear_archive/README.md`) preserves the exact official ASCII source, schema, metadata, parser and independent verifier. It accounts for **5,843 state rows and 3,558 distinct (A,Z) pairs**, including the free neutron at $Z=0$. Source columns, raw rows, uncertainties, unknown tokens and property-level systematics markers remain traceable. These are source-row counts, not a count of all presently observed isotopes or all usable material ingredients. A listed non-exist token remains such; a # mass marker is not a discovery classification.

The [evaluation paper](#) reports a 30 October 2020 data-availability boundary and a stated isomer scope. The downloaded ASCII's scientific cutoff is not independently established. Its server modification date and its one literal 2021 discovery-year field remain separate observations; neither silently changes the paper's boundary or proves when particular evidence reached the evaluators. A source-qualified identifier includes its edition, Z, A and original state index/suffix. This archive does not include every nuclear level, all 2026 observations or future nuclei. Its published-work CC BY 3.0 notice and the companion ASCII header's separate attribution situation remain documented.

The PDG 2026.0 identity archive (source file: `../data/particle_archive/README.md`) includes all **1,170 charge-state identity rows**, 450 selected particle/search identifier groups, 3,270 name/item rows, 1,341 mappings and 71 codebook rows from the pinned source, with source metadata. It preserves original IDs, aliases, nulls and ambiguous duplicate names. Its selection is complete for those tables/rows, not for the entire PDG publication or all particle properties. No discovery/existence assessments were performed. The archive omits available property measurements, masses/lifetimes, uncertainties, decays, references and other tables; a generic particle row or Monte Carlo identifier does not establish observation.

The official [PDG edition page](#) supplies the pinned database; the [published cutoff](#) is 15 January 2026. Release metadata and retrieval time have separate meanings. [API limitations](#) also preserve inaccessible or incomplete information. The [schema and license documentation](#) govern identity mappings and attributed CC BY 4.0 reuse. Searches and hypothetical models remain distinct from established observations.

An isotope identifier, particle name or model family supplies a reference address. Its combinations still need production/host, population, energy, lifetime, conservation/accounting, observation and useful-system descriptions. Free quarks, nuclei, their containing atoms and host-dependent excitations are not independent interchangeable chemical ingredients. The [CERN Standard Model account](#) supports those category and confinement distinctions. No finite archive establishes every future theoretical model, quantum state or physical entity.

Transferring composition bases between material classes

The new molecular, polymer and composite records retain the quantity actually described by their sources. A molecule's formula unit, a polymer's feed mass ratio and a filler's loading are different inputs to the elemental composition model. They cannot be inserted as interchangeable coordinates merely because each is expressed as a percentage.

For declared constituent phases a , let ϕ_a be their volume fractions and ρ_a their densities at the specified state and measurement convention. Where their volumes and masses can be consistently assigned, their mass fractions are $w_a = (\phi_a \rho_a) / (\sum_b \phi_b \rho_b)$. Include every relevant phase or state which contributes to that denominator. Void volume needs its own convention; it does not contribute an elemental atom population. If mass/volume assignment or density is unresolved, the conversion is unresolved.

Let $c_{e,a}$ be element e 's mass fraction within constituent a . With complete, compatible constituent inventories, the nominal mixture elemental mass fraction is $W_e = \sum_a w_a c_{e,a}$. If each constituent's effective molar mass for element e is $M_{e,a}$, its nominal elemental atom amount per unit total mass is $A_e = \sum_a (w_a c_{e,a} / M_{e,a})$, and $x_e = A_e / (\sum_j A_j)$. Each $M_{e,a}$ must account for that constituent's isotope/state population and the declared mass allocation convention. A constituent lacking element e contributes zero to A_e without needing $M_{e,a}$. Alternatively, convert separately identified isotope/state species to atom amounts before aggregating by element. When $W_e > 0$, the equivalent mixture-effective molar mass is $M_e = W_e / A_e$; a selected isotope mass or arithmetic mean of different constituent molar masses is not a general substitute. Unknown constituents, residuals or molar masses prevent a complete conversion; they are not zero by default. Binding-energy and metrology qualifications at extreme precision remain those described above. These equations are bookkeeping under the stated model, not new density or composition measurements.

CALCULATED illustrative conversion: for two ideal constituents with volume fractions (2/5,3/5) and densities in the ratio 2:1, the mass fractions are (4/7,3/7). No particular composite or measured density is asserted. This shows why a source's volume loading cannot automatically satisfy an atomic- or mass-fraction limitation.

For molecular solids, include the selected salt, hydration or solvation in formula-unit accounting, while keeping retained solvent/substrate/impurities in whole-product accounting. For polymers, repeat-unit accounting excludes finite-chain ends unless they are separately included. Monomer feed does not establish incorporated composition, sequence distribution or degradation products. Preserve these gaps when associating records with arbitrary rational composition addresses.

Specific references and conditional properties

The material reference entries (source file: `reference_entry_support.md`) supply individually identified published-literature records across metals/alloys, inorganic compounds, molecular materials, polymers and composites. They keep preparation and characterization details with their actual sources and list reproduction gaps. They do not represent applicant work, a newly discovered invention, a universal route, or a whole-product closed inventory where the source supports only principal-element or host-phase accounting. No field is automatically inherited as ESTABLISHED by a variant composition, isotope, process, structure or use.

A property scenario specifies the baseline material and state, controlled change, observable, units, conditions, expected mechanism, credible alternative, discriminating test and evidence status. Let $\Delta P = P_{\text{candidate}} - P_{\text{baseline}}$ for the same defined observable and test convention. Increased, decreased and unresolved responses can use declared thresholds $\Delta P \geq \tau$, $\Delta P \leq -\tau$ and $|\Delta P| < \tau$, with $\tau > 0$ justified by measurement resolution or model error. The threshold is not assumed known for an unmeasured material.

For a proposed measured comparison, separately predeclare a practical threshold τ and an uncertainty interval $[L_{\Delta}, U_{\Delta}]$ for the difference under a justified sampling/model method. An interval entirely above τ supports an increase; one entirely below $-\tau$ supports a decrease. An interval entirely within $[-\tau, \tau]$ can support no meaningful change under that declared criterion. All other intervals are inconclusive, including a point estimate beyond a threshold whose uncertainty crosses it. Threshold equality must follow the prespecified convention. Replicate independence, covariance, calibration and model uncertainty must be addressed; a confidence level or error model is not assumed validated merely because it is recorded. The five-family scenario cards (source file: `reference_entry_support.md`) keep null, contrary and inconclusive outcomes separate.

The contrary response is a separate conditional hypothesis with a mechanism and conditions that can distinguish it from the expected branch. It is not the simultaneous assignment of opposite properties to one specimen under identical conditions. A sign change in ΔP is distinct from an impossible negative absolute quantity. Strength, stiffness and torque-dependent geometry remain different quantities; electrical, thermal, optical and chemical observables retain their relevant frequencies, fields, temperatures, geometry and environments.

Where a predictive model is used, record its equations/version, parameter source, domain, boundary conditions, uncertainty and failure tests. Where these inputs are missing, the outcome is PROPOSED or UNSUPPORTED. A list of all possible signs or a declaration that unexpected effects are contemplated supplies research categories, not all future outcomes or a necessary inherent property.

Application integration and unresolved effect

The supplement and archives provide additional technical content for review. Files referenced only by repository links are not automatically included in an application. A final submitted provisional/PCT must carry the relied-on material in accepted application forms and preserve the exact submitted set. The filing memorandum (source file: `provisional_filing_strategy.md`) and current-status addendum (source file: `current_status_addendum.md`) distinguish preparation, publication and filing. [PCT description guidance](#) does not make a general incorporation statement a substitute for essential disclosure.

This extension does not add or alter the 199 candidate claims. Their textual support, physical enablement, novelty, inventorship and legal effect remain separate inquiries. [USPTO prior-art enablement guidance](#) and [novelty guidance](#) require assessment of actual teachings and later limitations. Experiments are not categorically required for every example, but mathematical addresses and dated labels cannot replace missing technical support. No application filing, universally prohibited later claim, comprehensive unexpected-property prediction or percentage of future patents blocked is asserted.

Source supported material reference entries

These entries provide concrete preparation and characterization evidence for the broad composition disclosure architecture. They describe published third-party work, not an applicant invention, performed applicant experiments, or a conclusion that all composition ranges are physically enabled. Their role is to make the architecture more specific while retaining the full requested scope and its unresolved technical gaps.

Current package evidence

The existing preparation framework in documents/provisional_application_draft.md, paragraphs [0038]-[0043], identifies the categories of information an entry needs. Paragraphs [0039]-[0040] list route classes rather than composition-specific recipes. Paragraph [0044] summarizes a real TiO₂(B) publication but explicitly leaves the complete methods to that source. Paragraph [0045] correctly limits what can be inferred from the host formula. Paragraphs [0062]-[0067] provide mechanism examples with proposed extrapolations, rather than an applicant-specific experimentally validated material series. Paragraph [0080] records unresolved full-scope preparation and property support.

The blueprint's Disclosure entry architecture already provides a useful field schema. Its existing MAT LIT 0001 is an attributed source locator, and says it is not a substitute recipe. The records below supply additional concrete fields and expose gaps that a composition vector alone cannot resolve. They do not convert a source summary into a demonstrated complete reproduction by the applicant.

Alloy reference family

Candidate identifiers: MAT-LIT-ALLOY-0001A and MAT-LIT-ALLOY-0001B. **Evidence:** ESTABLISHED third-party report; no applicant repetition. **Known published teaching:** Vaidya et al., *Scientific Reports* 7, 12293, September 25, 2017, DOI 10.1038/s41598-017-12551-9. [Primary article](#)

Field	Source supported detail
Nominal composition	Equiatomic CoCrFeNi and CoCrFeMnNi; respectively 25 and 20 atomic percent per principal element.
Starting materials	Solid constituent metals, 99.99 weight percent purity.
Preparation	Arc-melting chamber evacuated to 10 ⁻⁵ mbar and purged with purified argon; four or five remelts; homogenization at 1473 K for 50 hours.
Structure and composition checks	Single FCC phase reported; Cu K-alpha XRD, SEM with EBSD and EDS, and grain-boundary atom-probe analysis.
Atom-probe conditions	Tips at 60 K; laser pulses at 250 kHz and 20 pJ.
Conditional property evidence	Nickel grain-boundary diffusion in the five-component alloy exceeded the four-component alloy above approximately 800 K; the relationship reversed at lower temperatures.
Pinpoints	Methods; Results and Discussion, microstructure analysis; temperature dependence of Ni grain-boundary diffusion; Figures 2-4 and 6.
Not captured for reproduction	Charge masses, melt parameters, homogenization atmosphere/cooling, specimen-wide impurity uncertainty, and quantitative composition acceptance limits.

The original paper and its supplementary material remain the controlling technical sources for the cited experiment. The two records must retain separate identities. Neither may acquire an assumed composition

tolerance, impurity budget, cooling path, or property result through a default value in the proposed record generator.

Inference for the requested scope: a principal-element vector is only part of the material record. The exact rational targets (1/4, 1/4, 1/4, 1/4) and (1/5, 1/5, 1/5, 1/5, 1/5) are mathematical descriptions of nominal principal-element ratios. They do not assert that every atom in the isolated specimen belongs to those principal elements, or that measured ratios have infinite precision. Actual specimen-wide fractions require an assay and a denominator that accounts for the chosen impurity policy.

Proposed extension requiring independent support: a non-equiatomic composition, an added constituent, a different phase, an isotope-enriched feedstock, or a new annealing path becomes a new record linked to this family as background. Its preparation, measured identity, and property scenario remain PROPOSED or UNSUPPORTED until separately supported. The fifth-element comparison does not supply a rule for every sixth through 138th constituent.

Inorganic reference family

Candidate identifier: MAT-LIT-OXIDE-0001, extending the existing MAT LIT 0001. **Evidence:** ESTABLISHED third-party report; no applicant repetition. **Known published teaching:** Xiang et al., *Scientific Reports* 3, 1411, March 11, 2013, DOI 10.1038/srep01411. [Primary publisher PDF](#), [archived full article](#)

Field	Source supported detail
Identity	TiO ₂ (B) nanosheets with exposed (010) faces; Ti:O = 1:2 describes the oxide host.
Selected precursor route	1 mL TiCl ₄ , 1 mL deionized water, and 30 mL ethylene glycol in a 40 mL Teflon-lined autoclave; stir 30 seconds; seal; heat at 150 degrees C for six hours.
Isolation	Centrifuge; wash four times using water and ethanol.
Confirmation	Cu K-alpha XRD: wavelength 1.5418 angstrom, 40 kV, 200 mA, scan 6 degrees/minute; HRTEM at 200 kV. Raman and thermal analyses also inform identification.
Whole-product qualification	Surface ethylene glycol was estimated at about 22 weight percent. Water exposure could favor conversion to anatase.
Pinpoints	PDF page 2, Results and Figure 1; page 4, Discussion; page 5, Methods.
Not captured for reproduction	Precursor grades, cooling, centrifuge settings, wash volumes, drying/storage, yield, and quantitative impurity acceptance limits.

Accounting inference: this literature product should not be used as an established example of a whole isolated specimen containing only two elements. Its host formula describes one constituent phase. Retained molecular surface material introduces additional elemental inventory into whole-product accounting. The host, surface layer, and whole-product denominator must therefore have separate fields. The existing draft's host-formula arithmetic remains useful; a whole-product purity assumption would not be justified by that arithmetic.

Proposed extension requiring independent support: different Ti/O ratios, oxygen-vacancy concentrations, dopants, other polymorphs, different ligands, or trace constituent targets require distinct identities, routes, assays, and stability records. A source-backed preparation for this particular morphology does not transfer automatically to every point in the Ti/O composition simplex or to multicomponent oxide systems.

Molecular reference family

Identifier: MAT-LIT-MOLECULAR-0001. **Evidence:** ESTABLISHED third-party crystallization report; CALCULATED formula accounting; PROPOSED follow-up comparisons. No applicant experiment is reported.

Broadhurst et al., *IUCrJ* 8 (2021), 860-866, DOI 10.1107/S2052252521010101, report time-resolved carbamazepine crystallization. [Primary article, sections 2-3](#). Carbamazepine's formula is C₁₅H₁₂N₂O; its molecular identity requires the specified connectivity, not that formula alone. [NIST Chemistry WebBook identity](#).

Field	Attributed experimental detail
Feed and solvent	0.1125 g carbamazepine in 25.6257 g ethanol; water measured as 0.03%; gravity filtration.
Local preparation	Deposit 3 microlitres on a Quantifoil R2/2 grid plasma-treated for 45 s; hold at 298 K and 21% humidity for 20-180 s; pressure-assisted blotting and immediate vitrification in liquid ethane.
Characterization	CryoTEM and three-dimensional electron diffraction at 200 kV and 100 K.
Observations	At 30 s, dihydrate predominated and minor forms II, III and IV were identified. At 180 s, only dihydrate was observed. Powder X-ray diffraction detected dihydrate at both times.
Source limits	This is a grid crystallization study, not an isolated pure bulk batch of each minor phase; supporting information controls the detailed analysis.

CALCULATED accounting: excluding solvent, substrate, impurities and any bound water, a nominal carbamazepine formula unit has 30 atoms. In the ordered support (H,C,N,O), $x=(2/5, 1/2, 1/15, 1/30)$. Adding exactly two water molecules gives a nominal dihydrate unit C₁₅H₁₆N₂O₃ with 36 atoms and $x=(4/9, 5/12, 1/18, 1/12)$. Both sums are exactly one. These are formula-unit targets, not measured whole-product atomic fractions. The different unsolvated crystal forms retain the same nominal vector; a material address must also identify connectivity, crystal form, hydration and history.

Unresolved reproduction fields: quantitative phase fractions and their uncertainty, chosen-crystal sampling bias, a bulk isolation route for each minor form, residual solvent/impurity inventory, classification limits and preservation after warming. A detected crystallite does not establish a bulk phase fraction; a bulk nondetection does not establish absolute absence.

PROPOSED conditional scenario: compare the occurrence of non-dihydrate crystallites in a prespecified, consistently sampled region after short versus longer holds, under matched solution and environmental conditions. The reported time dependence motivates a decrease hypothesis. An increase or persistence must remain an unperformed alternative, with nucleation, transformation kinetics and sampling/measurement effects assessed separately. Do not assign an opposite strength, conductivity or solubility result from this study; those observables need their own specimens and protocols. Changing humidity, solvent or quench timing creates a new comparison and cannot silently inherit this source's result.

Coverage consequence: the record supplies molecular identity, a particular crystallization route and time-resolved structural observations. It does not supply every molecule sharing the element ratios, every polymorph, a closed whole-product impurity specification, or later manufacturing/use limitations. Retain this source as attributed background; no newly invented applicant material or universal molecular preparation rule is established.

Polymer reference family

Candidate identifiers: MAT-LIT-POLYMER-0001A (PLLA control) and MAT-LIT-POLYMER-0001B (PDLA/PLLA stereocomplex). **Status:** ESTABLISHED third-party reported preparation/measurements; no applicant repetition. Prepared 30 September 2026.

Primary publication: Su, Feng and Yu, *Polymers* 12 (2020), 2515, published 28 October 2020, DOI [10.3390/polym12112515](https://doi.org/10.3390/polym12112515). [Publisher article](#), [open archived article](#), [Europe PMC full-text XML](#). The article carries CC BY 4.0; no figures or full text are reproduced here.

Source data record

Field	Attributed detail
Identity/basis	PLLA versus PDLA/PLLA 1:1 polymer mass blend; PLLA grade 4032D.
Preparation	Vacuum-dry 40 °C overnight; melt-blend 190-220 °C, 15 rpm; cool 20 °C, granulate; hot-press 230 °C, 15 minutes; cool 20 °C.
DSC protocol	10 mg; nitrogen; 30-250 °C, 10 °C/min heating/cooling; 250 °C/3 minutes, then 30 °C/3 minutes; reported second heating.
Thermal result	Melting peaks: PLLA 171.2 °C; 1:1 blend 227.7 °C. Blend stereocomplex crystallinity: 43.7%.
Structure check	XRD: 2θ 5-30°, 5°/minute, 20 °C; stereocomplex peaks 11.9°, 20.7°, 24.0°.
Tensile protocol/result	20 °C, 20 mm/min, ISO-527-2, ≥5 tests. PLLA → blend: strength 52.98 → 83.70 MPa; modulus 1.13 → 2.04 GPa.
Pinpoints	Methods §§2.1-2.3; results §§3.1, 3.5; Figures 1, 8.

Accounting and material identity

CALCULATED ideal repeat-unit accounting: the constitutional repeat -O-CH(CH₃)-C(=O)- contains C₃H₄O₂. Its C/H/O atom fractions are 1/3, 4/9, and 2/9. The two stereochemical chain types share this bookkeeping vector, yet their sequence stereochemistry and crystal packing remain different record fields. [PubChem's L-lactide](#) and [D-lactide](#) records independently distinguish the monomer stereochemistry while giving the same C₆H₈O₄ formula.

The 1:1 feed ratio is neither a ratio of polymer molecules nor a whole-specimen elemental assay. Finite chains require end-group/initiator accounting; commercial material may also contain residual monomer, additives, moisture, or degradation products. No impurity budget, exact specimen-wide atom vector, or infinite measured precision is supplied by this calculation. Separate as-pressed film history from the thermal cycle imposed by DSC; a second-heating result must not automatically become an as-delivered service property. The reported crystallinity is not 100% of the specimen.

Reproduction gaps and source ambiguity

Our audit; unresolved: batch masses, vacuum pressure, extrusion residence/geometry, pressing pressure, cooling rate, optical purity, moisture/impurity limits, and raw uncertainty/calibration need completion. The PDLA Mn/PDI sentence is internally inconsistent; confirm its assignment independently. These gaps prevent this summary from being treated as a complete applicant reproduction. Keep the publication's route qualified rather than filling missing fields with assumed defaults.

Expected and contrary scenarios

Source interpretation: stereocomplex packing/physical junctions are associated with increased stiffness (§§3.3, 3.5). Physical junctions do not by themselves establish newly formed covalent branching.

PROPOSED matched-feed and lot test: hold feedstock lots and the 1:1 mass ratio fixed, then compare controlled thermal histories that favor retained crystallinity versus suppressed crystallization. Define an expected response as greater modulus when a measured stereocomplex fraction is greater, under a common temperature, strain-rate, geometry, conditioning, and molecular-weight protocol. Determine actual phase fractions before interpreting the mechanical result; matched feed does not establish matched final product composition.

PROPOSED contrary response: lower or unchanged modulus could occur if the chosen history instead limits crystallization, changes molecular weight, or introduces porosity/defects. Test those alternatives through structural, molecular-weight, and density evidence rather than declaring a contradiction from feed composition alone. A modulus increase also does not imply increased ductility or every form of toughness; choose separate response variables and decision thresholds before testing. Neither proposed comparison has been performed here.

This family supplies a concrete existing teaching for a particular polymer system. It does not establish new applicant subject matter, all-polymer preparation, properties for arbitrary elemental compositions, or a universal bar to later patents.

Composite reference family

Candidate: MAT-LIT-COMPOSITE-0001. **Classification:** ESTABLISHED third-party literature report; applicant reproduction NOT PERFORMED. **Reviewed:** September 30, 2026. This proposal extends the material-class support architecture with a composite example. It is neither an applicant invention nor evidence that the full proposed composition domain is enabled.

Bounded source record

Nguyen et al. (2017), *Carbon*, original NIST research. [NIST record](#); [author manuscript](#).

Field	Source-supported detail
Constituents/basis	Multiwall carbon nanotubes (MWCNTs), reported 0.72 mass% in amine-cured DGEBA epoxy; starting resin dispersion 1 mass%; polyoxypropylenetriamine hardener; no UV stabilizers.
Preparation	Magnetic stirring 1 h; room-temperature degassing 1 h; bar drawdown on PET; approximately 150 μm films. Toluene used, amount unspecified.
Cure	24 °C/45% RH, three days; postcure 110 °C, 4 h, circulating-air oven.
Exposure	295–400 nm, approximately 140 W/m ² ; 50 °C/75% RH. Characterization coupons 25 × 25 mm; release specimen approximately 78.5 cm ² .
Measurements	Mass change, ATR-FTIR (4 cm ⁻¹ , 128 scans), XPS, SEM/AFM/EFM; nanotube-rich surface network observed. Collector SEM detected no release after 4865 MJ/m ² .
Interface hypotheses	Anchoring, entanglement, residual-matrix bonding, and nanotube attractions proposed; individual contributions unverified.
Reproduction gaps	Batch masses, stirring speed, solvent quantity, nanotube lot/purity, quantitative release detection limit not supplied here.
Pinpoints	Manuscript §§2.1–2.3, lines 87–181; §§3.3–3.4, lines 282–490; PDF pages 5–9, 15–26.

Composition and identity interpretation

The entry generator must keep constituent loading, constituent identity, and whole-product elemental composition in separate fields. The percent-to-fraction conversion of a reported 0.72 mass% is $0.0072 = 9/1250$. That arithmetic does not make a rounded formulation statement an exact specimen assay. A record that uses 9/1250 should label it a representation of the reported loading, with measurement uncertainty unresolved, rather than an infinitely precise composition.

A nanotube mass fraction cannot be substituted for an atomic fraction or a volume fraction. Whole-product elemental fractions would require the matrix's constituent amounts and identities, reinforcement purity, residuals and contaminants, and an explicit denominator. A surface-sensitive analysis cannot establish those fractions throughout the specimen. Likewise, the record needs separate morphology and interface evidence; a nominal constituent vector does not encode dispersion, exposed-filler geometry, or adhesion strength.

The preparation fields above are useful source support, but they are not a complete executable protocol. Before an applicant repetition, obtain the missing formulation and instrument settings, specify material acceptance criteria and metrology uncertainty, and document the original source's supplementary methods. Do not silently supply those fields through generator defaults. Collector nondetection must retain its technique and collection conditions; it cannot become a numerical zero, a detection limit, or a general release guarantee.

Distinct published comparator

Zhao et al. (2023), *NanoImpact*, another original study: [NIST publication record and abstract](#). Its radiolabeled nanotube/epoxy specimens underwent UV plus water spray, followed by shaking or ultrasonication. Collected nanotube mass was reported as 0.23% of embedded nanotube mass during weathering; subsequent mechanical treatment released approximately 0.27% (mass/mass). The authors linked release to disruption of the surface network by spray. The 0.23% denominator is embedded nanotube mass, not whole-composite mass; the 0.27% denominator requires confirmation from the full methods. These are separate specimens and conditions, not measurements on MAT-LIT-COMPOSITE-0001. Only the official abstract was verified here; detailed formulation, spray settings, treatment durations, detection limits, and recovery corrections require the full methods before reproduction.

Proposed contrasting test and mechanisms

PROPOSED; no applicant result. Prepare independently documented replicate batches of the candidate and matched neat-matrix controls. Use a factorial comparison of UV exposure and calibrated water/mechanical stress, with temperature, humidity, specimen geometry, aging dose, and conditioning recorded. Include UV-only, stress-only, combined, and unexposed groups. Predefine the nanotube release measurement, blank correction, recovery, detection limit, sampling schedule, replicate count, and uncertainty. Record both released nanotube mass per initial embedded nanotube mass and per initial whole-specimen mass; retain particle morphology so nanotube-containing fragments are distinguished from isolated tubes.

The conditional response to test is whether a surface network persists under UV-only exposure but loses retention under the added stress. Retention could reflect geometrical interlocking and remaining matrix attachment; stress could remove the network or rupture its attachments. A release difference alone would not separate those mechanisms. Compare cross-sectional and surface microscopy, matrix chemical change, and independently measured interface response before and after treatment. If attachment and network geometry remain unchanged while release rises, investigate collection efficiency, degradation fragments, and other explanations before assigning a mechanism. These are testable alternatives, not promised outcomes.

Effect on the full-scope disclosure

This is a known published teaching and should be identified as background evidence. Its public provenance is established by the linked records; the exact legally operative publication date and claim comparison remain matters for the prior-art analysis. It supplies a concrete composite entry format and a bounded conditional property scenario. It does not provide preparation or property support for every matrix, filler, loading, interface, manufacturing route, isotope choice, or later use. Each extension needs its own identity, route, characterization, and support status; the universal defensive objective remains unresolved.

Evidence boundaries for record generation

Use the following field rules across every material family. These are proposed disclosure controls, not assertions about a particular source's successful experiments.

Field	Required rule	State when evidence is absent
Composition basis	State principal-element, host-phase, or whole-product accounting; retain exact target fractions separately from measured fractions and uncertainty.	Nominal target only.
Preparation	Preserve supported quantities and conditions together; record every unresolved critical step.	PROPOSED route or UNSUPPORTED feasibility, as applicable.
Identity	Establish which phase, morphology, connectivity, defect state, and sample history the record actually concerns.	Candidate identity only.
Characterization	Specify measurements able to distinguish realistic alternative products; add calibration, sampling, detection limits, and reference data.	Unconfirmed product or stated analytical gap.
Property	Bind a result to a named material, state, method, units, baseline, temperature, and other relevant conditions.	PROPOSED test or UNSUPPORTED prediction.
Variations	State which substitutions, interval regions, or process changes have independent supporting evidence.	No inherited evidence from a parent entry.
Provenance	Keep source teachings separate from applicant contributions and unperformed applicant work.	No applicant experiment, contribution, or inventorship established.

The broad range generator can reference these entry identifiers without representing any as a universal preparation algorithm. Generated records with no technical content must keep unresolved fields explicit. A process class, a property sign, or a list of opposite outcomes must never default to ESTABLISHED.

Specimen accounting declarations for proposed studies

These declarations specify what a new study must account for. They are PROPOSED boundaries and unresolved assay fields, rather than completed specimen certificates. Source nominal ratios remain those described above.

Family	Proposed physical boundary and accounting basis	Required residual or state treatment
Alloy	One isolated, conditioned coupon; principal-element target reported separately from a whole-coupon atomic/mass inventory.	Identify measured impurities, surface material, porosity and local segregation; report bounds/unknowns rather than setting every other element to zero.
Inorganic	Isolated nanosheet product after a named washing, drying and storage procedure; report oxide-host and whole-product inventories separately.	Include retained ligand, hydration, solvent and non-host phases; specify their mass or atom bases and assay uncertainty.
Molecular	Selected crystallites in defined grid regions at the observation state; formula-unit calculations remain separate from grid/whole-deposit inventory.	Specify hydration/solvation, unclassified crystals and retained solvent; exclude the support only under an explicit crystal-only boundary.
Polymer	Conditioned molded specimen from identified feed lots and thermal history; feed masses and incorporated product composition are separate records.	Include end groups, additives, residuals, degradation and porosity; do not equate matched feed with an established matched final composition.
Composite	Isolated cured film plus separately tracked collected material; define dry/wet conditioning and cured-film loading.	Track resin, hardener, nanotubes, retained solvent, voids, surface enrichment and collection losses; embedded-nanotube and whole-film mass denominators remain distinct.

For a particle-system extension, declare whether the counted population consists of atoms, ions, nuclei, electrons, other particles or host-dependent excitations. Specify the boundary, host, population/occupation convention, normalization, observation interval and lifetime treatment. Do not reuse a chemical elemental fraction as a particle-population fraction without a justified conversion. Hypothetical entities remain unresolved identities and never receive measured population or persistence by default.

Proposed property scenario cards

All five cards below are unperformed applicant-study designs. Published observations remain ESTABLISHED third-party background, the accounting examples remain CALCULATED, and the new comparison outcomes/mechanisms remain PROPOSED or UNSUPPORTED as stated. These cards do not supply new measured signs, thresholds, confidence levels or acceptance limits. Complete the unresolved protocol fields before conducting or interpreting a study.

For each card, preserve individual specimen and batch identifiers, raw data, calibration records, excluded observations and the reason for each exclusion. Use an initial pilot with independently prepared specimens to estimate variability and then justify the replicate count and sampling frame for the intended precision. A repeated measurement or multiple crystals from one preparation is not automatically an independent preparation. Randomize or counterbalance assignments where appropriate and retain failed preparations. The comparison must use the same defined observable and units in both branches.

Predeclare a meaningful-change threshold $\tau > 0$ and a justified uncertainty interval $[L_Delta, U_Delta]$ for the signed difference Δ . Use an interval wholly above τ for increase, wholly below $-\tau$ for decrease, wholly within $[-\tau, \tau]$ for no meaningful change, and otherwise inconclusive. A null finding is not a negative-sign response; an inconclusive interval is not evidence of equivalence. Set confidence/model conventions and threshold equality in the protocol. Account for shared calibration, covariance, clustering and multiple comparisons when applicable. No value of τ or uncertainty is fabricated here.

SC-ALLOY-01: transport under specified temperature conditions

Protocol field	Proposed definition and unresolved input
Baseline and change	Compare MAT-LIT-ALLOY-0001A with 0001B at each nominated common temperature, with independently prepared, characterized coupons. Composition and temperature changes have separate identities.
Observable and units	Within a common B-type analysis, use $\Delta = \ln(P_B/P_A)$, dimensionless, for $P = s \Delta D_{gb}$ in m^3/s , requiring $P_A > 0$ and $P_B > 0$. A nondetection or interval reaching zero remains censored/bounded; it cannot be inserted as zero in this logarithm. Within a C-type analysis, use the separately identified D_{gb} in m^2/s . Do not compare a triple product directly with a diffusion coefficient.
Necessary state record	Grain structure, phase, composition/residuals, tracer boundary, annealing temperature/time, penetration-depth calibration and fitting regime. A conversion between quantities must retain the chosen segregation/width model and its uncertainty.
Background and branches	The source reports a temperature-dependent ordering. The new experiment must determine its own sign at each stated temperature; no ordering transfers to arbitrary temperatures, added elements or hypothetical labels.
Discriminating evidence	Raw tracer profiles, justified regime/fitting limits, fit uncertainty and independently assessed microstructure. A response does not by itself establish the proposed mechanism.
Unresolved fields	Exact nominated temperatures, tracer inventory/procedure, profile sampling, model parameters, preparation tolerances, replicate count, tau and uncertainty method.

The distinction between P and D_{gb} follows the [primary article's B/C regime analysis and Figure 6](#). Their interchange requires an additional physical model, rather than a unit relabeling.

SC-OXIDE-01: phase response to a defined exposure

Protocol field	Proposed definition and unresolved input
Baseline and change	Split a characterized starting product into matched aliquots; compare a declared water-exposure history with a declared control history. Match initial host/surface-material accounting.
Observable and units	$\Delta = f_{\text{anatase,exposed}} - f_{\text{anatase,control}}$, dimensionless, using phase fractions under one validated quantitative convention. Separately report $\text{TiO}_2(\text{B})$, other phases and any unquantified material.
Necessary state record	Initial phase fractions, surface ligand/hydration, specimen mass, exposure temperature/time and storage/measurement history. Retained material can change the whole-product elemental inventory.
Expected and other branches	A conversion concern motivates a proposed increase. Persistence is a null alternative. A decrease requires its own mechanism or remains UNSUPPORTED; it is not supplied by listing the opposite sign.
Discriminating evidence	Calibrated quantitative phase analysis and an independent structural check, with detection/quantification limits and preparation replicates. Distinguish crystallite detection from bulk phase fraction.
Unresolved fields	Exposure schedule, complete phase-quantification method, standards/model, residual treatment, replicate count, tau and uncertainty method.

SC-MOLECULAR-01: sampled crystallite occurrence over time

Protocol field	Proposed definition and unresolved input
Baseline and change	Compare short and longer holds from matched solution lots and preparation conditions, with separately prepared grids. Fix region-selection rules before inspection.
Observable and denominator	$p_{\text{minor}} = (\text{classified non-dihydrate crystallites}) / (\text{all evaluated crystallites})$; $\Delta = p_{\text{minor, long}} - p_{\text{minor, short}}$. Report unclassified counts separately and bound their possible effect. This is a count fraction, not a bulk mass/volume phase fraction.
Necessary state record	Hold/quench timing, local environment, grid region, hydration/solvation and classification conditions. Matching the initial solution does not establish matching final deposits.
Expected and other branches	Source observations motivate a proposed decrease. An increase is a separate unperformed kinetic/sampling hypothesis; unchanged frequency and inconclusive evidence are separate outcomes.
Discriminating evidence	Prespecified sampled regions, crystal classifications and reliability checks, independent grid/preparation replicates, and an uncertainty method that accounts for within-grid clustering and unclassified crystals.
Unresolved fields	Representative sampling frame, target times, classification error, preservation after quench, repeat count, tau and interval method. No representative population frequency is inferred from the existing selected-crystal observations.

SC-POLYMER-01: thermal history with matched feed and lot

Protocol field	Proposed definition and unresolved input
Baseline and change	Compare two declared thermal histories for matched 1:1 PDLA/PLLA feed masses and lots. Final specimen composition and molecular-weight distributions require independent checks.
Observable and units	$\Delta = E_{\text{history A}} - E_{\text{history B}}$, in GPa, under the same specimen geometry, conditioning and tensile/modulus fitting protocol. Specify gauge length or local strain measurement and fitting interval; crosshead speed alone is not strain rate.
Necessary state record	As-processed stereocomplex/homocrystal fractions, porosity, chain/end-group/degradation evidence and retained residuals. Second-heating DSC does not automatically characterize the original tensile-test specimen state.
Expected and other branches	More stereocomplex crystallinity motivates a proposed modulus increase; degradation, voids or inadequate crystallization may produce a lower or unresolved result. Neither branch is an applicant measurement or a guaranteed causal relationship.
Discriminating evidence	Independent processing batches, phase measurements in the tensile-test state, local strain/modulus analysis and chain/porosity controls. Keep the source control/blend comparison separate from this fixed-feed comparison.
Unresolved fields	Source molecular-weight ambiguity, exact histories/press pressure/cooling, purity/lot checks, specimen protocol, replicate count, tau and uncertainty method.

SC-COMPOSITE-01: release under matched exposure and collection

Protocol field	Proposed definition and unresolved input
Baseline and change	Within one characterized formulation, compare UV-only treatment with a declared combined exposure/stress treatment using matched conditioning, dose and collection. The two published studies do not establish matched specimens.
Observable and denominator	$r_{CNT} = (\text{recovered released nanotube mass}) / (\text{initial embedded nanotube mass})$; Delta = $r_{combined} - r_{UV}$. Separately report release relative to initial whole-film mass, collection recovery and possible unrecovered material.
Necessary state record	Cured loading, formulation, retained solvent, film geometry, dose, water/mechanical conditions, collector boundaries and time. Distinguish isolated nanotubes from nanotube-containing fragments.
Expected and other branches	Added exposure/stress motivates a proposed increase. Unchanged release remains a separate outcome. A negative branch needs an independently supported mechanism or remains UNSUPPORTED; a nondetection is a bounded observation, not an exact zero.
Discriminating evidence	Blanks, recovery controls, limits of detection/quantification, mass balance and released-material morphology with independently prepared films. Preserve isotope-label/assay conventions if used.
Unresolved fields	Complete matched cure/exposure/stress protocol, initial embedded mass, recovery/calibration, collection geometry, repeat count, tau and uncertainty method. The comparator abstract's 0.27% denominator is not silently assumed.

Property scenarios and unresolved breadth

The user's request to describe expected and contrary responses can be implemented as a conditional scenario structure: specified baseline, mechanistic rationale, candidate change, defined test, decision threshold, and unresolved alternatives. A scenario containing both signs remains a proposal until evidence distinguishes them. Storing both possible responses does not establish that either occurs for an arbitrary element combination, and cannot establish a quantitative property threshold by itself.

For these reference families, a future applicant study would need specimen provenance, raw composition and structural records, and a separately defined property protocol. No study has been run here. Changes at 10^{-19} fractional abundance, all constituent counts through 138, hypothetical elements, isotope-state availability, and particle containment remain outside the support provided by these five published families.

These references are existing public teachings. Republishing them adds an attributed record to this package; it does not make the underlying work newly disclosed by the applicant. No patent-status search, freedom-to-operate determination, claim-specific anticipation opinion, full-scope enablement conclusion, or universal patent bar is asserted. The records strengthen the evidence architecture while leaving the requested full scope intact for further technical and legal assessment.

Specific claim fields and later claim comparisons

The original claims 176, 177 and 178 identify molecular, polymer and composite fields. They all depend on claim 1, including its closed atomic/mass inventory and exact zero outside that inventory. The following is an evidence map, not a novelty or complete support finding.

Original candidate	New entry fields available for review	Required features still unresolved
Claim 176	Molecular identity, hydration, time-dependent solid forms and a local crystallization procedure.	Whole-specimen inventory and exact-zero residuals; relevant stereochemistry/solvation and quantitative phase selection; independently assessed full claim scope.
Claim 177	Named chain stereochemistry, ideal repeat unit, feed mass ratio, processing and measured phase/property comparisons.	Incorporated whole-product elemental ratios, end groups, chain/sequence distributions and network completeness; source ambiguity and exact-zero residuals.
Claim 178	Identified matrix/reinforcement, loading basis, film preparation, interface questions and conditional release measurements.	Complete constituent inventory, retained loading, directional/volume accounting, quantified interface/dispersion and exact-zero residuals.
Claims 197-199	Baselines, conditional alternatives, proposed distinguishing measurements and local evidence labels.	Applicant-specific conception, novelty/eligibility and support for the claimed methods; physical outcomes remain unperformed where stated.

These entries also make the blueprint's hypothetical later-claim tests more concrete:

Later limitation scenario	Features to compare	Missing features and possible argument
A material and preparation matching one identified entry	Match identity, state, quantity basis and the linked preparation steps together in the controlling source.	A section 102 argument is a candidate only after verifying every limitation, enablement and a qualifying date. A summary's omitted critical steps must be checked in the source and relevant skilled-person knowledge.
The same elements/nominal proportions with a selected solid form or architecture	Separately locate the claimed form, chain stereochemistry or interface geometry.	An element vector alone leaves these features open. An observed minor crystal does not establish an isolated high-purity bulk product. Section 103 would need particular teachings, reasoning and a supported expectation of success.
A selected composition or loading from a very large catalog	Identify the particular jointly taught embodiment and denominator, including any necessarily inherent features or limited combination that a skilled person would at once envisage, rather than independently selecting unrelated menus.	A named species can be disclosed in a large list; an unspecified selection is not automatically anticipated. A narrower range, critical value or morphology needs its own comparison.
A later manufacturing method or use	Compare every step, operating condition and functional limitation with the particular source.	Product identity alone does not establish additional method or use limitations. Assess whether each limitation is expressly taught or necessarily inherent, and separately assess any supported section 103 rationale. No blanket rejection follows.

This map does not assess a real later patent claim or assign a blocked percentage. Anticipation and obviousness require their distinct, fact-specific analyses. [USPTO MPEP 2131](#), [MPEP 2143](#).

Content to include in an actual filing

If these teachings are relied upon, include their necessary identity, preparation, conditions, equations and drawings directly in the accepted description or clearly identified description annex. Identify the exact pages supporting each relied-on feature. Preserve the uploaded and accepted versions. A repository link, current script or live supplement does not automatically become application content. If earlier provisional support is sought, the relied-on content must be present in that earlier filing; later additions do not retroactively enter it. PCT missing-part incorporation has specific prior-application, document and timing conditions and is not a general incorporation of a live repository. [WIPO ISPE 4.12 and 4.26-4.27](#), [PCT Rule 20.6](#), [USPTO provisional guidance](#).

The applicant's actual technical contribution remains TECH-001. These attributed published experiments do not become applicant discoveries by being summarized here. Inventorship and any new claimed contribution must be established separately; experiments are not categorically required in every case. [USPTO MPEP 2109](#).

Current publication and filing status

Status checked: 30 September 2026 UTC. This addendum supersedes historical preparation-stage statements about publication in the original memorandum. It does not change the original PDFs or their preparation dates.

The research package is publicly accessible in [RobertMLayne/all-materials-patent](#). The recorded observation (source file: `../data/publication_observation.json`) identifies the exact main-branch version and the observed public-access time. The original documents were prepared on 29 September 2026. Neither that preparation date nor a commit's creation time is certified as the earliest public-availability date.

No patent filing, official receipt, accepted priority claim, fee payment, or micro-entity certification is asserted. The package's public disclosure and any future application filing have different legal functions and dates. Public disclosure can potentially qualify as a publicly accessible reference under the applicable rules; it supplies only the actual teachings of the publicly accessible version. It does not create a provisional filing date, establish that every candidate material is enabled, or certify a universal bar. [USPTO MPEP 2152.02](#).

An actual disclosure chronology is necessary before pursuing the applicant's own patent rights. The U.S. inventor-originated disclosure exceptions require their conditions to be established, and foreign rights may be affected even if a U.S. exception applies. A later provisional does not restore a universally prepublication position. This note records the issue without assuming a particular grace-period exception applies. [USPTO provisional application guidance](#).

The original 199 candidate claims remain unassessed for claim-specific novelty, and their physical enablement remains unresolved. Four are independent claims: 1, 139, 186 and 194. With all 199 retained at U.S. national stage, the illustrative micro excess-claim charges are 179 x USD40 plus 1 x USD120, totaling **USD7,280 before other charges**. The official schedule retrieved on 30 September 2026 lists USD40 under 1.492(e) and USD120 under 1.492(d). This does not apply to the provisional, is not a complete filing quote, and depends on valid micro eligibility and the fee schedule applicable when payment is made. [USPTO fee schedule](#), [37 CFR1.492\(d\)-\(e\)](#).

The intended PCT filing should contain its essential technical disclosure in accepted application contents. Merely referring to this repository, scripts, or a provisional is not a general substitute. Ordinary PCT requests designate the states bound on the international filing date; later national or regional phase decisions select where protection is pursued. [PCT Rule4.9](#), [PCT ISPE4.26-4.27](#).

This supplement is drafting and evidence work. No filing action or legal-coverage conclusion is reported.

Full-scope objective audit

Prepared 30 September 2026 UTC. This audit preserves the user's full requested outcome: a comprehensive provisional/PCT materials disclosure spanning elemental supports and proportion ranges through 138 labels, isotopes and known/hypothetical particles, expected and contrary properties, and an intended bar to all or nearly all later materials patents. It does not redefine that outcome as successful numerical enumeration or document preparation.

The original working draft has 80 numbered paragraphs and 199 candidate claims. The extension adds an unrestricted-domain and physical-accounting supplement (source file: technical_scope_supplement.md), specific literature records (source file: reference_entry_support.md), dated nuclear (source file: ../data/nuclear_archive/README.md) and particle (source file: ../data/particle_archive/README.md) identity archives, and a current publication-status addendum (source file: current_status_addendum.md). Original claims, paragraph numbering and PDFs are retained. There is no official filing receipt or claimed applicant experiment in this package.

Full requirement audit

User requirement	Current evidence	Completion status and missing evidence
Choose the broad all-element option without arbitrary exclusions	Paragraphs [0004], [0008], [0015]-[0022]; 138 ordered label rows; claims 1-143	Organizing scope present. Z=1...118 are recognized; Z=119...138 are hypothetical placeholders. No evidence establishes the placeholders' existence, chemistry, availability or a maximum natural element count of 138.
Include presently unprotected materials, known or otherwise	Broad candidate domain and attributed existing teachings	No blanket unprotected status established. There is no global patent-status, ownership or freedom-to-operate inventory. A known composition, unknown candidate and legally unprotected material are different descriptions; their statuses cannot be inferred from the element vocabulary.
Every support from one element and every 2-138-element combination	Nonempty support definition and exact count formulas; count candidates through 138 and unary claim 139	Mathematical addressing present. All physical combinations are not supported by preparation/structure teachings. Counts above 118 require hypothetical constituents. Permutations do not create distinct support sets, while physical arrangement still requires a separate record.
Every ordinary percent cell and complementary proportion, including 1-2%/98-99% and 50/50	[0023]-[0027], [0069]-[0072]; claims 152-163; decimal and exact-rational examples	Numerical definitions present. Complement coordinates are correlated. They are not proof of product composition, exact purity, a phase or manufacturing precision.
Precision through the stated $10^{-16}\%$... $10^{-15}\%$ interval and down to fraction $1/10^{19}$, or lower practical limits	Exact percent/fraction conversion; the optional $D=10^{19}$ grid; unrestricted positive rational implementation and below-grid examples	Exact rational computation now extends below the former executable floor. Real targets remain symbolic. There is no verified universal practical concentration floor, manufacturing tolerance or instrument operating at the grid precision. Fixed specimens need integer counts.
All normalized ranges jointly across every support, without a later meaningful restriction/broadening	Arbitrary rational interval feasibility, arbitrary-q lattice witnesses, positive-support boundary handling and primitive enumeration	Mathematical scope present; legal outcome unproved. Particular later ranges, critical values, structures and species need their own disclosure comparison. A broad menu does not automatically teach every selected combination with the required specificity.
A sufficiently detailed comprehensive provisional disclosure	Substantial original description, drawings, claims, supplemental definitions, dated entity data and specific reference records	Review package developed; full-scope sufficiency unproved. No applicant technical contribution or composition-specific preparation/generalization supports every indexed material. References retain known reproduction gaps and third-party provenance.
All isotope and nuclear-state possibilities	[0046]-[0050]; claims 169-173; complete import of 5,843 state rows/3,558 (A,Z) pairs from the selected NUBASE ASCII	Dated source archive populated. Includes free neutron and uncertain/estimated/source-qualified entries; not all current/future nuclei or every excited level. The ASCII's independent scientific cutoff is not established. Joint count compatibility does not prove a state exists or is preparable.

User requirement	Current evidence	Completion status and missing evidence
All proven and theoretical subatomic particles and combinations	Original 17 elementary categories; record-method claims 194-196; PDG 2026.0 selected identity/name/search/mapping/code tables	Specified identity selections populated. Zero physical-existence assessments performed. Properties, production, confinement, occupations and lifetimes are not supplied by this identity export. Future models/entities and every quantum state are not a closed, verified set.
Predict expected material properties, every alteration and opposite/unexpected effects	[0057]-[0068], claims 197-199; baseline/change/alternative/test framework and attributed mechanism examples	Scenario format and some existing teachings present. No composition-indexed universal prediction set or applicant result exists. Observable, state, conditions, mechanism, magnitude, uncertainty and distinguishing evidence remain absent for most candidates. Mutually conditional alternatives are not simultaneous verified outcomes or necessary inherency.
Claim all ranges despite exceeding 20 claims	All 199 candidates are printed, with per-claim textual locations and novelty-unassessed status	Requested large claim annex present. Claim numbers do not multiply technical support. No claim-specific novelty, written-description, full-scope enablement, eligibility or inventorship conclusion has been reached.
Provisional followed by a worldwide PCT incorporating it, initially U.S. and later other member countries	Filing memorandum (source file: provisional_filing_strategy.md), accepted-content and ordinary timing discussion	Procedure analyzed; no filing performed. Essential content must enter the actual accepted application. The PCT request ordinarily designates all treaty states bound at its filing date; later phase choices determine where protection is pursued. The proposed U.S.-only-then-add-states sequence is not the ordinary designation mechanism.
Fees calculated using micro-entity status	Conditional fee illustrations and eligibility distinctions in the memorandum	Illustrations present; actual quote/eligibility unproved. No certification, applicant-specific qualifying facts or submitted chargeable-sheet count is established. U.S. micro treatment does not itself establish WIPO's separate reduction. Later claim/page/search/phase fees depend on the actual filing.
Publication adequate to create defensive prior art	Public original repository version recorded in publication observation (source file: ../data/publication_observation.json); status addendum distinguishes preparation and filing	Observed original public availability recorded. Earliest public availability is not certified; new extension publication needs its own final version/access evidence. Public access is not an official provisional/PCT filing or an earlier statutory patent-document reference date.
Prohibitive prior art against all or nearly all future combinations, properties, manufacturing methods and uses	Broad scope plus actual, individually attributed teachings	Full requested outcome unproved. No qualifying reference is shown to disclose and enable every later limitation; no universal obviousness, date/exception or worldwide conclusion is supported. No blocked percentage is assigned.

Evidence present, and what it proves

The unrestricted model (source file: `../range_model/unrestricted_composition_model.json`), examples (source file: `../range_model/unrestricted_numeric_examples.json`), implementation (source file: `../range_model/unrestricted_compositions.py`) and recorded report (source file: `../range_model/unrestricted_verification_report.json`) use exact fractions without a chosen positive floor. All support cardinalities are checked mathematically. Closed-box/open-simplex feasibility and arbitrary-denominator count feasibility are distinct. A fixed N-atom realization requires the primitive elemental denominator to divide N; isotope/state partitions require each joint sub-count to be integral as well.

The implementation report records 11 bounded groups. Separate independent checks passed 1,812 positive-box cases, 2,400 lattice boxes, 14,609 primitive-vector/inventory cases, 1,805 joint-partition cases, eight invalid-partition cases and 65 explicit uniform-subset cases. Those oracles use vertices, separator positions, independent common-denominator counts and explicit subsets. They support the algorithms and stated proofs. They do not experimentally test the indexed materials or exhaust an infinite domain.

The nuclear verifier accounts for every preserved source row and reconstructs raw columns; its deliberate-defect cases reject lost rows, altered identities/status and misread bounds. The portable package can reconstruct the optional normalized table in memory without a new network download. This proves fidelity to one source file, not the existence or availability of every listed state. The paper's 30 October 2020 availability statement, the ASCII server modification date, its literal 2021 discovery-year field and the retrieval date retain separate meanings.

The particle verifier reports all selected source rows/columns match the pinned PDG SQLite source. Original nulls, aliases, search groups and ambiguous duplicate names remain. The export contains 1,170 charge-state identity rows and complete declared supporting selections, rather than all measurements or the entire publication. [PDG's API scope and limitations](#) remain applicable. Neither the PART code nor the database release status is a discovery assessment.

The reference entries (source file: `reference_entry_support.md`) develop separately identified records across all five material families in the blueprint. They supply particular source-backed preparation/characterization details, quantity-basis conversions and conditional comparisons, and explicitly retain missing reproduction fields. The new molecular/polymer/composite records are mapped to the relevant original claim fields without treating them as evidence for claim 1's exact-zero whole-product inventory. They remain attributed prior publications; adding them to this package does not create an applicant experiment, newly conceived invention or universal preparation rule.

Technical fields still missing over the requested physical domain

Material class	Important unresolved fields for a selected candidate
Metals/alloys	Actual feed purity and charge, phase relations, atmosphere, volatility/immiscibility, consolidation, thermal/mechanical history, specimen assay and property protocol.
Inorganic compounds/ceramics	Precursors, charge/site and defect constraints, reaction path, phases/polymorphs, crystallization/conditioning, isolation, purity and structure confirmation.
Molecular materials	Connectivity, stereochemistry, counterions/solvation, reaction or assembly, purification and chemical identity; overall element percentages do not supply a molecule.
Polymers	Repeat units, incorporated composition, sequence/tacticity, distributions, end groups, branching/crosslinks, preparation and conditioning.
Composites/mixtures	Constituent identities, geometry, interfaces, dispersion, phase/void fractions, assembly, local/bulk denominator and property direction.
Nuclear/particle systems	Actual species/state, existence evidence, production, energy/occupation convention, host/confinement, interactions, lifetime, decay inventory, observation and a physically meaningful system/use.

Across all classes, identify sample size/region, measurement uncertainty, detection/quantification capability, impurities and unknown residuals. A zero in a mathematical target is not a measured absence. A nominal host or principal-element ratio is not automatically a closed whole-product inventory. Charge, site occupancy, stability, processing history and isotope partitioning must not be defaulted from a normalization witness.

Property work needs a baseline, observable, conditions, comparison threshold, supported model or actual data and a credible alternative mechanism. Increased/decreased/unchanged categories leave infinitely many magnitudes, transitions, anisotropies, histories and uses unspecified. A source-supported response for one material family is not evidence for every sixth through 138th constituent or hypothetical nuclide.

Checks against illustrative later claims

These are architecture checks, not opinions on an identified patent. Applicable prior-art dates and exceptions are not assumed established.

Later claimed feature	Disclosed feature and outstanding comparison
A specifically identified material made by an adequate route	The reference records contain particular source teachings and named preparation/characterization fields. Compare every actual later limitation with the relevant source and its supporting knowledge; the generic framework supplies no universal route.
Same elements with different proportions or crystal structure	Rational targets and structure fields can address that candidate. Its particular phase, occupied sites, defects and making conditions still require disclosure; another phase is not established by a matching bulk vector.
A composition selected from the enormous indexed list	A unique numerical address is present. Multiple selections from composition, structure and process menus are not automatically a specifically taught embodiment. The required enabling and novelty comparison remains open.
A later manufacturing method, use or unexpected property	Conditional property scenarios and route classes provide questions and background. Newly claimed steps, conditions or use features must be matched separately. An inherent property requires necessity in the relevant identical material, rather than the listing of possible opposite outcomes.

The [USPTO anticipation guidance](#) and [inherency guidance](#) govern those legal comparisons. A section 103 argument needs the relevant actual teachings and supported selection/combination rationale; no universal argument follows from these numerical checks.

Legal and filing requirements still missing

The intended prior-art effect requires claim-by-claim comparisons of actual teachings, support, applicable dates and statutory exceptions. [USPTO novelty guidance](#) does not make generic menus automatically disclose every species or narrower range. [Prior-art enablement guidance](#) distinguishes teaching how the subject matter can be carried out from merely naming it. Experiments are not categorically required for every example, and absence of applicant experiments alone is not the decisive gap; missing technical direction across this physical scope is.

Record the actual human technical contribution and inventors, applicant/entitlement information, relevant disclosure/filing history, jurisdictional requirements, fee eligibility and chosen receiving Office/search authority. No values are invented. Prepare and verify the accepted submitted description, claims, drawings, abstract and relied-on annexes. [PCT content guidance](#) and [Rule 4.9](#) govern essential disclosure/designations; repository links and general incorporation language are not filing receipts or a cure for absent support. Unity, meaningful-search and national/regional requirements remain separate assessments.

The original publication observation establishes an observed public version by its stated observation time, not a backdated preparation date. Any final extension publication must identify its own exact contents and public-access evidence. A future provisional does not automatically restore a universally prepublication position; the actual disclosure chronology and jurisdiction-specific consequences need review. No successful filing, fee payment or certification is reported here.

Completion finding and next evidence

The full objective remains unproved. The extension improves exact numerical coverage, finite/joint inventory accounting, dated entity identities and individually supported reference records. It leaves the requested universal physical and legal outcome intact as an unresolved requirement.

The next evidence-bearing work is to identify the applicant's actual contribution; complete selected material/state/preparation/property records; justify any supported generalizations across the intended class; compare particular later limitations to qualifying teachings; and assemble an accepted filing set if filing is separately authorized. Keep unsupported regions and unsuccessful results explicit. Do not substitute document counts, checksums, a green workflow, archive-row counts, extra claims or possible opposite responses for those technical/legal conclusions.

Per-claim page and textual-support map

Every claim remains a candidate. Textual locations do not establish physical enablement, novelty, eligibility, unity or inventorship. Original Part A page numbers are used below.

Claim	Original paragraph locations	Part A claim page	Review status
1	[0010] (A 2), [0011] (A 2), [0014] (A 2), [0015] (A 3), [0016] (A 3), [0080] (A 11)	A 13	Textual map; novelty, legal support and relevant physical scope remain unresolved.
2	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 13	Textual map; novelty, legal support and relevant physical scope remain unresolved.
3	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 13	Textual map; novelty, legal support and relevant physical scope remain unresolved.
4	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 13	Textual map; novelty, legal support and relevant physical scope remain unresolved.
5	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 13	Textual map; novelty, legal support and relevant physical scope remain unresolved.
6	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 13	Textual map; novelty, legal support and relevant physical scope remain unresolved.
7	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 13	Textual map; novelty, legal support and relevant physical scope remain unresolved.
8	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 13	Textual map; novelty, legal support and relevant physical scope remain unresolved.
9	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 13	Textual map; novelty, legal support and relevant physical scope remain unresolved.
10	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 13	Textual map; novelty, legal support and relevant physical scope remain unresolved.
11	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 13	Textual map; novelty, legal support and relevant physical scope remain unresolved.
12	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 13	Textual map; novelty, legal support and relevant physical scope remain unresolved.
13	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 13	Textual map; novelty, legal support and relevant physical scope remain unresolved.
14	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 13	Textual map; novelty, legal support and relevant physical scope remain unresolved.
15	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 13	Textual map; novelty, legal support and relevant physical scope remain unresolved.
16	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 13	Textual map; novelty, legal support and relevant physical scope remain unresolved.
17	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 13	Textual map; novelty, legal support and relevant physical scope remain unresolved.

Claim	Original paragraph locations	Part A claim page	Review status
18	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 13	Textual map; novelty, legal support and relevant physical scope remain unresolved.
19	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 13	Textual map; novelty, legal support and relevant physical scope remain unresolved.
20	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 13	Textual map; novelty, legal support and relevant physical scope remain unresolved.
21	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 13	Textual map; novelty, legal support and relevant physical scope remain unresolved.
22	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 13	Textual map; novelty, legal support and relevant physical scope remain unresolved.
23	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 13	Textual map; novelty, legal support and relevant physical scope remain unresolved.
24	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 13	Textual map; novelty, legal support and relevant physical scope remain unresolved.
25	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 14	Textual map; novelty, legal support and relevant physical scope remain unresolved.
26	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 14	Textual map; novelty, legal support and relevant physical scope remain unresolved.
27	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 14	Textual map; novelty, legal support and relevant physical scope remain unresolved.
28	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 14	Textual map; novelty, legal support and relevant physical scope remain unresolved.
29	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 14	Textual map; novelty, legal support and relevant physical scope remain unresolved.
30	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 14	Textual map; novelty, legal support and relevant physical scope remain unresolved.
31	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 14	Textual map; novelty, legal support and relevant physical scope remain unresolved.
32	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 14	Textual map; novelty, legal support and relevant physical scope remain unresolved.
33	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 14	Textual map; novelty, legal support and relevant physical scope remain unresolved.
34	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 14	Textual map; novelty, legal support and relevant physical scope remain unresolved.
35	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 14	Textual map; novelty, legal support and relevant physical scope remain unresolved.
36	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 14	Textual map; novelty, legal support and relevant physical scope remain unresolved.

Claim	Original paragraph locations	Part A claim page	Review status
37	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 14	Textual map; novelty, legal support and relevant physical scope remain unresolved.
38	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 14	Textual map; novelty, legal support and relevant physical scope remain unresolved.
39	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 14	Textual map; novelty, legal support and relevant physical scope remain unresolved.
40	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 14	Textual map; novelty, legal support and relevant physical scope remain unresolved.
41	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 14	Textual map; novelty, legal support and relevant physical scope remain unresolved.
42	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 14	Textual map; novelty, legal support and relevant physical scope remain unresolved.
43	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 14	Textual map; novelty, legal support and relevant physical scope remain unresolved.
44	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 14	Textual map; novelty, legal support and relevant physical scope remain unresolved.
45	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 14	Textual map; novelty, legal support and relevant physical scope remain unresolved.
46	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 14	Textual map; novelty, legal support and relevant physical scope remain unresolved.
47	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 14	Textual map; novelty, legal support and relevant physical scope remain unresolved.
48	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 14	Textual map; novelty, legal support and relevant physical scope remain unresolved.
49	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 14	Textual map; novelty, legal support and relevant physical scope remain unresolved.
50	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 14	Textual map; novelty, legal support and relevant physical scope remain unresolved.
51	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 14	Textual map; novelty, legal support and relevant physical scope remain unresolved.
52	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 14	Textual map; novelty, legal support and relevant physical scope remain unresolved.
53	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 14	Textual map; novelty, legal support and relevant physical scope remain unresolved.
54	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 14	Textual map; novelty, legal support and relevant physical scope remain unresolved.
55	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 14	Textual map; novelty, legal support and relevant physical scope remain unresolved.

Claim	Original paragraph locations	Part A claim page	Review status
56	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 14	Textual map; novelty, legal support and relevant physical scope remain unresolved.
57	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 14	Textual map; novelty, legal support and relevant physical scope remain unresolved.
58	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 15	Textual map; novelty, legal support and relevant physical scope remain unresolved.
59	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 15	Textual map; novelty, legal support and relevant physical scope remain unresolved.
60	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 15	Textual map; novelty, legal support and relevant physical scope remain unresolved.
61	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 15	Textual map; novelty, legal support and relevant physical scope remain unresolved.
62	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 15	Textual map; novelty, legal support and relevant physical scope remain unresolved.
63	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 15	Textual map; novelty, legal support and relevant physical scope remain unresolved.
64	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 15	Textual map; novelty, legal support and relevant physical scope remain unresolved.
65	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 15	Textual map; novelty, legal support and relevant physical scope remain unresolved.
66	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 15	Textual map; novelty, legal support and relevant physical scope remain unresolved.
67	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 15	Textual map; novelty, legal support and relevant physical scope remain unresolved.
68	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 15	Textual map; novelty, legal support and relevant physical scope remain unresolved.
69	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 15	Textual map; novelty, legal support and relevant physical scope remain unresolved.
70	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 15	Textual map; novelty, legal support and relevant physical scope remain unresolved.
71	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 15	Textual map; novelty, legal support and relevant physical scope remain unresolved.
72	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 15	Textual map; novelty, legal support and relevant physical scope remain unresolved.
73	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 15	Textual map; novelty, legal support and relevant physical scope remain unresolved.
74	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 15	Textual map; novelty, legal support and relevant physical scope remain unresolved.

Claim	Original paragraph locations	Part A claim page	Review status
75	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 15	Textual map; novelty, legal support and relevant physical scope remain unresolved.
76	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 15	Textual map; novelty, legal support and relevant physical scope remain unresolved.
77	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 15	Textual map; novelty, legal support and relevant physical scope remain unresolved.
78	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 15	Textual map; novelty, legal support and relevant physical scope remain unresolved.
79	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 15	Textual map; novelty, legal support and relevant physical scope remain unresolved.
80	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 15	Textual map; novelty, legal support and relevant physical scope remain unresolved.
81	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 15	Textual map; novelty, legal support and relevant physical scope remain unresolved.
82	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 15	Textual map; novelty, legal support and relevant physical scope remain unresolved.
83	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 15	Textual map; novelty, legal support and relevant physical scope remain unresolved.
84	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 15	Textual map; novelty, legal support and relevant physical scope remain unresolved.
85	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 15	Textual map; novelty, legal support and relevant physical scope remain unresolved.
86	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 15	Textual map; novelty, legal support and relevant physical scope remain unresolved.
87	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 15	Textual map; novelty, legal support and relevant physical scope remain unresolved.
88	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 15	Textual map; novelty, legal support and relevant physical scope remain unresolved.
89	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 15	Textual map; novelty, legal support and relevant physical scope remain unresolved.
90	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 15	Textual map; novelty, legal support and relevant physical scope remain unresolved.
91	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 16	Textual map; novelty, legal support and relevant physical scope remain unresolved.
92	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 16	Textual map; novelty, legal support and relevant physical scope remain unresolved.
93	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 16	Textual map; novelty, legal support and relevant physical scope remain unresolved.

Claim	Original paragraph locations	Part A claim page	Review status
94	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 16	Textual map; novelty, legal support and relevant physical scope remain unresolved.
95	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 16	Textual map; novelty, legal support and relevant physical scope remain unresolved.
96	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 16	Textual map; novelty, legal support and relevant physical scope remain unresolved.
97	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 16	Textual map; novelty, legal support and relevant physical scope remain unresolved.
98	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 16	Textual map; novelty, legal support and relevant physical scope remain unresolved.
99	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 16	Textual map; novelty, legal support and relevant physical scope remain unresolved.
100	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 16	Textual map; novelty, legal support and relevant physical scope remain unresolved.
101	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 16	Textual map; novelty, legal support and relevant physical scope remain unresolved.
102	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 16	Textual map; novelty, legal support and relevant physical scope remain unresolved.
103	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 16	Textual map; novelty, legal support and relevant physical scope remain unresolved.
104	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 16	Textual map; novelty, legal support and relevant physical scope remain unresolved.
105	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 16	Textual map; novelty, legal support and relevant physical scope remain unresolved.
106	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 16	Textual map; novelty, legal support and relevant physical scope remain unresolved.
107	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 16	Textual map; novelty, legal support and relevant physical scope remain unresolved.
108	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 16	Textual map; novelty, legal support and relevant physical scope remain unresolved.
109	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 16	Textual map; novelty, legal support and relevant physical scope remain unresolved.
110	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 16	Textual map; novelty, legal support and relevant physical scope remain unresolved.
111	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 16	Textual map; novelty, legal support and relevant physical scope remain unresolved.
112	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 16	Textual map; novelty, legal support and relevant physical scope remain unresolved.

Claim	Original paragraph locations	Part A claim page	Review status
113	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 16	Textual map; novelty, legal support and relevant physical scope remain unresolved.
114	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 16	Textual map; novelty, legal support and relevant physical scope remain unresolved.
115	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 16	Textual map; novelty, legal support and relevant physical scope remain unresolved.
116	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 16	Textual map; novelty, legal support and relevant physical scope remain unresolved.
117	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 16	Textual map; novelty, legal support and relevant physical scope remain unresolved.
118	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 16	Textual map; novelty, legal support and relevant physical scope remain unresolved.
119	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 16	Textual map; novelty, legal support and relevant physical scope remain unresolved.
120	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 16	Textual map; novelty, legal support and relevant physical scope remain unresolved.
121	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 16	Textual map; novelty, legal support and relevant physical scope remain unresolved.
122	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 16	Textual map; novelty, legal support and relevant physical scope remain unresolved.
123	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 16	Textual map; novelty, legal support and relevant physical scope remain unresolved.
124	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 17	Textual map; novelty, legal support and relevant physical scope remain unresolved.
125	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 17	Textual map; novelty, legal support and relevant physical scope remain unresolved.
126	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 17	Textual map; novelty, legal support and relevant physical scope remain unresolved.
127	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 17	Textual map; novelty, legal support and relevant physical scope remain unresolved.
128	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 17	Textual map; novelty, legal support and relevant physical scope remain unresolved.
129	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 17	Textual map; novelty, legal support and relevant physical scope remain unresolved.
130	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 17	Textual map; novelty, legal support and relevant physical scope remain unresolved.
131	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 17	Textual map; novelty, legal support and relevant physical scope remain unresolved.

Claim	Original paragraph locations	Part A claim page	Review status
132	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 17	Textual map; novelty, legal support and relevant physical scope remain unresolved.
133	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 17	Textual map; novelty, legal support and relevant physical scope remain unresolved.
134	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 17	Textual map; novelty, legal support and relevant physical scope remain unresolved.
135	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 17	Textual map; novelty, legal support and relevant physical scope remain unresolved.
136	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 17	Textual map; novelty, legal support and relevant physical scope remain unresolved.
137	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 17	Textual map; novelty, legal support and relevant physical scope remain unresolved.
138	[0015] (A 3), [0016] (A 3), [0022] (A 3), [0073] (A 9)	A 17	Textual map; novelty, legal support and relevant physical scope remain unresolved.
139	[0013] (A 2), [0015] (A 3), [0016] (A 3)	A 17	Textual map; novelty, legal support and relevant physical scope remain unresolved.
140	[0004] (A 1), [0015] (A 3)	A 17	Textual map; novelty, legal support and relevant physical scope remain unresolved.
141	[0004] (A 1), [0015] (A 3), [0041] (A 5)	A 17	Textual map; novelty, legal support and relevant physical scope remain unresolved.
142	[0015] (A 3), [0016] (A 3), [0022] (A 3)	A 17	Textual map; novelty, legal support and relevant physical scope remain unresolved.
143	[0015] (A 3), [0019] (A 3), [0022] (A 3)	A 17	Textual map; novelty, legal support and relevant physical scope remain unresolved.
144	[0010] (A 2)	A 17	Textual map; novelty, legal support and relevant physical scope remain unresolved.
145	[0010] (A 2)	A 17	Textual map; novelty, legal support and relevant physical scope remain unresolved.
146	[0011] (A 2)	A 17	Textual map; novelty, legal support and relevant physical scope remain unresolved.
147	[0011] (A 2), [0047] (A 6)	A 17	Textual map; novelty, legal support and relevant physical scope remain unresolved.
148	[0017] (A 3), [0073] (A 9), [0074] (A 9)	A 17	Textual map; novelty, legal support and relevant physical scope remain unresolved.
149	[0017] (A 3), [0074] (A 9)	A 17	Textual map; novelty, legal support and relevant physical scope remain unresolved.
150	[0018] (A 3), [0031] (A 4)	A 18	Textual map; novelty, legal support and relevant physical scope remain unresolved.

Claim	Original paragraph locations	Part A claim page	Review status
151	[0019] (A 3), [0072] (A 9)	A 18	Textual map; novelty, legal support and relevant physical scope remain unresolved.
152	[0029] (A 4), [0030] (A 4)	A 18	Textual map; novelty, legal support and relevant physical scope remain unresolved.
153	[0023] (A 3)	A 18	Textual map; novelty, legal support and relevant physical scope remain unresolved.
154	[0023] (A 3), [0024] (A 3)	A 18	Textual map; novelty, legal support and relevant physical scope remain unresolved.
155	[0024] (A 3), [0029] (A 4), [0069] (A 9)	A 18	Textual map; novelty, legal support and relevant physical scope remain unresolved.
156	[0025] (A 4), [0026] (A 4), [0070] (A 9)	A 18	Textual map; novelty, legal support and relevant physical scope remain unresolved.
157	[0070] (A 9)	A 18	Textual map; novelty, legal support and relevant physical scope remain unresolved.
158	[0070] (A 9)	A 18	Textual map; novelty, legal support and relevant physical scope remain unresolved.
159	[0070] (A 9)	A 18	Textual map; novelty, legal support and relevant physical scope remain unresolved.
160	[0070] (A 9)	A 18	Textual map; novelty, legal support and relevant physical scope remain unresolved.
161	[0027] (A 4)	A 18	Textual map; novelty, legal support and relevant physical scope remain unresolved.
162	[0017] (A 3), [0071] (A 9)	A 18	Textual map; novelty, legal support and relevant physical scope remain unresolved.
163	[0072] (A 9)	A 18	Textual map; novelty, legal support and relevant physical scope remain unresolved.
164	[0018] (A 3), [0020] (A 3)	A 18	Textual map; novelty, legal support and relevant physical scope remain unresolved.
165	[0020] (A 3)	A 18	Textual map; novelty, legal support and relevant physical scope remain unresolved.
166	[0028] (A 4)	A 18	Textual map; novelty, legal support and relevant physical scope remain unresolved.
167	[0027] (A 4)	A 18	Textual map; novelty, legal support and relevant physical scope remain unresolved.
168	[0016] (A 3), [0027] (A 4)	A 18	Textual map; novelty, legal support and relevant physical scope remain unresolved.
169	[0046] (A 6), [0047] (A 6)	A 18	Textual map; novelty, legal support and relevant physical scope remain unresolved.

Claim	Original paragraph locations	Part A claim page	Review status
170	[0046] (A 6), [0047] (A 6)	A 18	Textual map; novelty, legal support and relevant physical scope remain unresolved.
171	[0047] (A 6)	A 18	Textual map; novelty, legal support and relevant physical scope remain unresolved.
172	[0048] (A 6), [0049] (A 6)	A 19	Textual map; novelty, legal support and relevant physical scope remain unresolved.
173	[0050] (A 6)	A 19	Textual map; novelty, legal support and relevant physical scope remain unresolved.
174	[0033] (A 5), [0034] (A 5)	A 19	Textual map; novelty, legal support and relevant physical scope remain unresolved.
175	[0035] (A 5)	A 19	Textual map; novelty, legal support and relevant physical scope remain unresolved.
176	[0035] (A 5)	A 19	Textual map; novelty, legal support and relevant physical scope remain unresolved.
177	[0036] (A 5)	A 19	Textual map; novelty, legal support and relevant physical scope remain unresolved.
178	[0036] (A 5)	A 19	Textual map; novelty, legal support and relevant physical scope remain unresolved.
179	[0013] (A 2), [0037] (A 5)	A 19	Textual map; novelty, legal support and relevant physical scope remain unresolved.
180	[0037] (A 5)	A 19	Textual map; novelty, legal support and relevant physical scope remain unresolved.
181	[0033] (A 5), [0034] (A 5)	A 19	Textual map; novelty, legal support and relevant physical scope remain unresolved.
182	[0033] (A 5), [0036] (A 5)	A 19	Textual map; novelty, legal support and relevant physical scope remain unresolved.
183	[0012] (A 2), [0036] (A 5)	A 19	Textual map; novelty, legal support and relevant physical scope remain unresolved.
184	[0042] (A 6), [0043] (A 6)	A 19	Textual map; novelty, legal support and relevant physical scope remain unresolved.
185	[0013] (A 2), [0038] (A 5)	A 19	Textual map; novelty, legal support and relevant physical scope remain unresolved.
186	[0005] (A 1), [0007] (A 2), [0073] (A 9), [0074] (A 9), [0075] (A 9)	A 19	Textual map; novelty, legal support and relevant physical scope remain unresolved.
187	[0015] (A 3), [0073] (A 9)	A 19	Textual map; novelty, legal support and relevant physical scope remain unresolved.
188	[0017] (A 3), [0073] (A 9), [0074] (A 9)	A 19	Textual map; novelty, legal support and relevant physical scope remain unresolved.

Claim	Original paragraph locations	Part A claim page	Review status
189	[0018] (A 3), [0070] (A 9), [0073] (A 9)	A 19	Textual map; novelty, legal support and relevant physical scope remain unresolved.
190	[0023] (A 3), [0029] (A 4), [0032] (A 4), [0074] (A 9)	A 20	Textual map; novelty, legal support and relevant physical scope remain unresolved.
191	[0029] (A 4), [0030] (A 4)	A 20	Textual map; novelty, legal support and relevant physical scope remain unresolved.
192	[0030] (A 4)	A 20	Textual map; novelty, legal support and relevant physical scope remain unresolved.
193	[0031] (A 4)	A 20	Textual map; novelty, legal support and relevant physical scope remain unresolved.
194	[0051] (A 7), [0052] (A 7), [0053] (A 7), [0056] (A 7)	A 20	Textual map; novelty, legal support and relevant physical scope remain unresolved.
195	[0054] (A 7)	A 20	Textual map; novelty, legal support and relevant physical scope remain unresolved.
196	[0055] (A 7), [0056] (A 7)	A 20	Textual map; novelty, legal support and relevant physical scope remain unresolved.
197	[0057] (A 7), [0058] (A 7), [0059] (A 8), [0060] (A 8)	A 20	Textual map; novelty, legal support and relevant physical scope remain unresolved.
198	[0057] (A 7), [0061] (A 8), [0062] (A 8), [0063] (A 8), [0064] (A 8), [0065] (A 8), [0066] (A 8), [0067] (A 8), [0068] (A 9)	A 20	Textual map; novelty, legal support and relevant physical scope remain unresolved.
199	[0005] (A 1), [0041] (A 5), [0042] (A 6), [0075] (A 9)	A 20	Textual map; novelty, legal support and relevant physical scope remain unresolved.

New partial evidence

Part B supplies exact-domain and accounting qualifications. Part C adds the five individually identified literature families and SC-ALLOY-01, SC-OXIDE-01, SC-MOLECULAR-01, SC-POLYMER-01 and SC-COMPOSITE-01. Claims 176-178 still inherit claim 1's closed whole-product inventory and exact zero outside it. Formula, host, feed and filler-loading accounting does not establish those limitations. Claims 197-199 receive further proposed scenario/evidence fields without new applicant measurements or validated universal mechanisms.

Actual physical preparations, unknown recipe fields, assay limits and human contribution must be evaluated separately. No map row promotes a candidate to allowed, enabled or novelty-cleared status.